

International Rocket Engineering Competition

2027 Design, Test, Evaluation Guide

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Verify that this is the latest revision before use.

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1.0 INTRODUCTION

The Experimental Sounding Rocket Association (ESRA) hosts and supports the International Rocketry Engineering Competition (IREC), a week-long series of events which will set the background and provide structure for the world's largest university rocket engineering competition. This new host event continues the IREC legacy of inspiring student design teams from across the country and around the world.

1.1 BACKGROUND

The smoke and fire, noise, high speeds, and sleek aerodynamics of rocketry encourage students to pursue science, technology, and mathematics-based careers. They have Rocket Fever! and competition motivates them to extend themselves beyond the classroom to design and build the rockets themselves. These students also learn to work as a team, solving real world problems under the same pressures they'll experience in their future careers.

ESRA held the first annual IREC in 2006. The competition achieved international status in 2011 when Canadian and Brazilian universities arrived. These schools have since been joined by others from every continent except Antarctica. The competition has increased in size every year since 2013, becoming the largest known collegiate level rocket engineering competition in the world. Attendance in 2016 included as many as 600 participants – including faculty, family, and friends of students from over 50 colleges and universities. The next year marked the start of a new era with the inaugural Spaceport America Cup. Over 1,100 students and representatives from 22 industry partners participated in an academic conference, rocket and payload engineering competitions, and non-competing demonstration flight tests. IREC continues to grow with nearly 200 teams applying for admission to the competition in 2023 with 121 teams and over 2,000 students from around the world making the trip to Spaceport America in June of 2023 for the competition.

For 2025 and 2026, the competition transitioned to Midland, Texas with the strategic objective of inspiring students to elevate their engineering capabilities. This included the introduction of a 45,000-foot two-stage rocket category and a 100,000-foot waiver to push the boundaries of altitude achievement. The 2026 IREC marked a record-breaking year, drawing over 2,000 students to the event and engaging nearly 8,000 participants representing nearly 150 team rocketry programs.

1.2 PURPOSE AND SCOPE

D&T-1.2.1 This document defines the minimum design, test, and evaluation criteria the event organizers expect teams to meet before launching at the IREC. The event organizers use these criteria to promote flight safety. Departures from the guidance this document provides may negatively impact an offending team's score and flight status, depending on the degree of severity. The foundational, qualifying criteria for the IREC are contained in the IREC Rules & Requirement document.

D&T-1.2.2 This document builds upon:

- Tripoli Rocketry Association (TRA) Unified Safety Code (USC)
- National Fire Protection Association (NFPA) Code for High Power Rocketry (NFPA 1127). While NFPA 1127, Section 1.3.3, exempts colleges and universities from its contents, these documents remain excellent supplemental resources for student researchers to learn more about best practices adopted by the amateur high-power rocketry community.
- ESRA's observations on student launch initiatives.
- Feedback received from student teams.

D&T-1.2.3 If any IREC team is unclear about competition rules and/or requirements, or has a situation not specifically addressed by the rules, they should contact ESRA by posting a message in the "dteg-questions" channel on Discord. Teams are encouraged to ask about both requirement interpretation and the spirit and intent of the rules and the DTEG.

1.3 CONVENTION AND NOTATION

The following definitions differentiate between requirements and other statements. The degree to which a team satisfies the spirit and intent of these statements will guide the competition officials' decisions on a project's flight status and overall score at the IREC.

Shall This is the only verb used to denote mandatory requirements. Failure to satisfy the spirit and intent of a mandatory requirement will always affect a project's score and flight status.

May This verb signifies that teams are allowed, but not required, to exercise the associated action.

Should This verb is used for stating non-mandatory goals. Failure to satisfy the spirit and intent of a non-mandatory goal may affect a project's score and flight status, depending on design implementation and the team's ability to provide thorough documentary evidence of their due diligence on-demand.

Will This verb is used for stating facts and declarations of purpose. The authors use these statements to clarify the spirit and intent of requirements and goals.

1.4 FLIGHT STATUS

Flight status refers to the granting of permission to attempt flight, and the provisions under which that permission remains valid. A project's flight status may be either nominal, provisional, or denied.

Nominal A project assigned nominal flight status meets or exceeds the minimum expectations of this document and reveals no obvious flight safety concerns during flight safety review at the IREC.

Provisional A project assigned provisional flight status generally meets the minimum expectations of this document but reveals flight safety concerns during flight safety review at the IREC which may be mitigated by field modification or by adjusting launch environment constraints. Launch may occur only when the prescribed provisions are met.

Denied A project assigned Denied flight status has failed to meet the minimum expectations of this document or the Tripoli Unified Safety Code or reveals unmitigated flight safety concerns. Competition officials reserve the right to deny flight status to any project deemed Denied.

1.5 LAUNCH VEHICLE VERSUS PAYLOAD

An effort is made throughout this document to differentiate between launch vehicle and payload associated systems. Unless otherwise stated, requirements referring only to the launch vehicle do not apply to payloads and vice versa.

1.6 REVISIONS

The IREC DTEG may require revision periodically (from time to time or from one competition to the next), based on the lessons learned by both host organizations and the participants.

- Major revisions will be accomplished by complete document reissue.
- "Real world events" may require smaller revisions to this document in the months leading up to a competition.
- Revisions will be reflected in updates to the document's effective date.
- The authority to approve and issue revised versions of this document rests with ESRA.

2.0 REFERENCE DOCUMENTATION

The following documents include standards, guidelines, schedules, or required standard forms. The documents listed in this section are either applicable to the extent specified in this document or contain reference information useful in the application of this document.

Table 2-1 - Applicable and Reference Documents

DOCUMENT	FILE LOCATION
Location of Current IREC documents	https://www.esrarocket.org/documents-and-forms
IREC Rules & Requirements Document	https://www.esrarocket.org/s/IREC-Rules-and-Requirements-Document-2026-v10.pdf
IREC Integrated Master Schedule Document	https://www.esrarocket.org/s/2027-IREC-IMS-REV-P1.pdf
IREC Frequency Student Band Plan	https://www.esrarocket.org/s/IREC-2026-Frequency-Student-Band-Plan-Rev-D-041426.pdf
IREC Range Standard Operating Procedures[1]	https://www.esrarocket.org/s/irec_range_standard_operating_procedures_2025-a.pdf
IREC Live Rocket Video Challenge	https://www.esrarocket.org/media-challenges
IREC Entry Form & Progress Report	https://www.esrarocket.org/2027-irec-team-application
TRA Unified Safety Code	https://www.tripoli.org/content.aspx?page_id=22&club_id=795696&module_id=520420
NFPA 1127: Code for High-Power Rocketry	https://www.nfpa.org/codes-and-standards/all-codes-and-standards/list-of-codes-and-standards/detail?code=1127
14 CFR, Part 1, 1.1 General Definitions	http://www.ecfr.gov/cgi-bin/text-idx?SID=795aaa37494b6c99641135267af8161e&mc=true&node=se14.1.1_11&rgn=div8

[1] The 2027 IREC Range Standard Operating Procedures is being updated and will be made available before the 2027 IREC.

DOCUMENT	FILE LOCATION
14 CFR, Part 101, Subpart C, 101.22 Definitions	http://www.ecfr.gov/cgi-bin/text-idx?SID=795aaa37494b6c99641135267af8161e&mc=true&node=se14.2.101_122&rgn=div8
47 CFR, Part 97, Amateur Radio Service	https://www.ecfr.gov/current/title-47/chapter-I/subchapter-D/part-97

3.0 HIGH POWER CERTIFIED FLYER OF RECORD

3.1 TRIPOLI FLIGHT REQUIREMENTS FOR ALL COMPETITION CATEGORIES

D&T-3.1.1 Every team shall engage a TRA Adult Flier, having a certification level appropriate for the motor being flown at IREC, to serve as their Flyer of Record (FoR). A Level 2 TRA Certification is required for K or L impulse motors or a combination of motors having a combined total impulse of less than 5,120 Ns. A Level 3 TRA Certification is required for M, N or O impulse motors or a combination of motors having a total impulse greater than 5,120 Ns. Any one of the following options shall satisfy this FoR requirement.

- Recruit an L2 or L3 TRA-certified Adult Flier with significant rocketry experience to work closely with the team and then attend IREC as the FoR. Student teams are encouraged to compensate or cover the travel expenses of their FoR.
- If a team member (a student) meets the requirement of Section 3.2, the student team member can be the FoR.[2]
- Teams who have no access to local or regional Tripoli prefectures should first attempt a search using the ESRA Mentor/FoR database, social media, forums, and other web-based sites (Facebook, Discord, *etc*.) to find a suitable FoR. If that search is unsuccessful, teams should contact the Tripoli Rocketry Association's Outreach Committee for assistance. (https://www.tripoli.org/outreach)

D&T-3.1.2 The FoR shall be present for the Fight Readiness Review (Horseshoe Pavilion), launch preparations, the Final RSO Check (immediately prior to entering queue), and for pad activities.

D&T-3.1.2 It is recommended that motors or energetic materials only be possessed or handled by TRA members (including Team members) with appropriate certification.

[2] ESRA reserves the right to review the qualifications and experience of a student FoR.

3.2 TEAM MEMBERS

D&T-3.2.1 Teams flying in the Solid or Hybrid/Liquid categories, or the Demonstration category if using solid or hybrid/liquid motors, shall meet all TRA launch safety requirements as codified in the then-current Tripoli Rocketry Association Unified Safety Code.

D&T-3.2.2 All team members who will be onsite at the launch site shall hold a current TRA membership. TRA Student Membership satisfies this requirement.

D&T-3.2.3 A maximum of 5 student team members may be on the pad loading team, although pad loading teams in the two-stage and hybrid/liquid areas may be larger at the discretion of the pad manager.

- The team Mentor may be present at the pads and does not count in the team limit in 3.2.3.
- The FoR shall be present at the pads and does not count in the team limit in 3.2.3.

D&T-3.2.4 The recovery team shall contain a minimum of 2 and a maximum of 4 team members per 100 lbs. of recovery equipment, including the dry weight of the rocket. The Mentor and/or FoR may accompany the recovery team and are not included in the team member count.

4.0 FREQUENCY MANAGEMENT AND GPS TRACKING REQUIREMENTS

4.1 STUDENT RADIO TRANSMISSIONS

D&T-4.1.1 All student radio transmissions shall meet the requirements of this section and the IREC 2027 Student Band Plan (SBP).

D&T-4.1.2 The Student Band Plan (SBP) addresses the transmission characteristics for COTS GPS systems, SRAD transmissions and Video transmissions, and may address allowable transmission bands, number of transmitters, frequency selection, forward RF power, transmission bandwidth (FWHM), and spurious transmissions and harmonics.

- The SBP may be updated from time to time or from one competition to the next, based on lessons learned by both ESRA and the participants. Revisions will be reflected in updates to the document's effective date. The authority to approve and issue revised versions of the SBP rests with ESRA.
- Revisions of the SBP are not considered as revisions of this DTEG document but are incorporated into the DTEG by reference.
- The SBP is not an authorizing document. Participants are still fully responsible for complying with all applicable FCC regulations, regardless of anything stated in the plan.

4.2 COTS GPS TRACKING SOLUTIONS

D&T-4.2.1 Each independently recovered section of the rocket (i.e., the rocket itself, a two-stage booster, or a deployed payload) shall include one Primary COTS GPS tracking solution. Recommended (but not required) COTS GPS trackers are listed in the table below. IREC-specific operating modes and frequencies are listed in the SBP.

Table 4-1 - Recommended COTS GPS Tracking Solutions

VENDOR	PRODUCT		WEBSITE
Altus Metrum	TeleGPS, TeleMega, *etc.*	70 cm	https://altusmetrum.org/
Big Red Bee	70cm GPS/APRS and 70cm & 33cm LoRa	70 & 33 cm	http://www.bigredbee.com/
Featherweight	Featherweight GPS & Featherweight Swift	33 cm	https://www.featherweightaltimeters.com/
Entacore	Aim XTRA	33 cm	https://entacore.com/
Silicdyne	Fluctus & Reperix	33 cm	https://silicdyne.net/products/
Missileworks	RRC4/T4 250mw	33 cm	https://www.missileworks.com/

Multitronix	Kate	33 cm	https://www.multitronix.com/

D&T-4.2.2 COTS GPS systems from other suppliers may be used if they meet the requirements of this section, including the requirements listed in the IREC SBP. The use of APRS protocol, as described in Section 0, is recommended.

D&T-4.2.3 COTS GPS tracking solutions shall be switched and powered independently of any flight computer.

- COTS flight computers with integrated GPS tracking satisfy the requirements of both COTS GPS Tracking and Recovery Electronics and are exempt from the (D&T-4.2.3) independent switch and power requirements.

4.3 FREQUENCY MANAGEMENT

D&T-4.3.1 ESRA will be responsible for frequency management during the IREC.

- ESRA will designate a specific frequency range (*i.e.*, COTS frequencies) within the 900 MHz (33 cm) and 440 MHz (70 cm) bands for the exclusive use of Primary COTS GPS trackers. Secondary COTS GPS trackers and all other SRAD transmitters must operate outside of the COTS frequencies.[3]
- ESRA may designate specific frequency ranges for other purposes (e.g., SRAD transmissions or Live Video), as specified in the SBP.
- ESRA may choose to assign transmitter frequencies either in advance of, or at, the competition, or alternatively, not assign frequencies. These assignment procedures are described in the SBP.[4]
- All RF sources must be documented in the required periodic Progress Reports, or as specified by ESRA, and must meet SBP requirements.

D&T-4.3.2 For 2027, the anticipated frequency assignment procedures for Primary COTS GPS and SRAD transmitters are as follows:

- Teams can elect to use a Primary COTS GPS tracker, operating within the COTS GPS tracker range of the 440 MHz (70 cm) band, for any independently recovered section of the rocket. Teams selecting this frequency band, and using a fixed-frequency transmitter, will be assigned one (or more) tracker frequencies by ESRA prior to the competition.
- Teams can instead elect to use a Primary COTS GPS tracker, operating within the COTS GPS range of the 900 MHz (33 cm) band, for any independently recovered section of the rocket. Teams selecting this frequency band, and using a fixed-frequency transmitter, will

[3] Note that the Missileworks and Multitronix COTS GPS units have some or all of their operation outside of the Primary COTS frequency range. See the SBP for details.

[4] In 2026, Primary COTS GPS and SRAD transmitter frequencies in the 440MHz band were preassigned. Assignments for Primary COTS GPS and SRAD frequencies in the 900MHz band were made at the time of the Final Flight Safety Review. Assignments for transmitters in the 1.2, 2.4, 3.3 and 5.8GHz bands were made at the time of the Final Flight Safety Review on a case-by-case basis.

be assigned one (or more) tracker frequencies at the time of the Final Flight Safety Review. Teams will surrender the use of these frequencies after completing the Post Flight Data Recovery process.

- Team can instead elect to use a Primary COTS GPS tracker, operating in frequency bands other than 440 and 900MHz bands, for any independently recovered section of the rocket. Teams should not anticipate frequency assignments; however, ESRA may direct the use of a specific frequency at the time of the Final Flight Safety Review in an effort to assist teams in deconflicting these transmissions.
- Teams operating Secondary COTS or SRAD GPS systems, or other SRAD transmitters, operating in the 440MHz (70 cm) and 900MHz (33 cm) ranges, and using a fixed-frequency transmitter, will receive frequency assignments at the time of the Final Flight Safety Review. These transmitters shall use frequencies outside of the COTS GPS tracker ranges in the 440 MHz (70 cm) and 900 MHz (33 cm) bands, within ranges as specified in the SBP.[5]
- Teams operating Secondary COTS or SRAD GPS systems, or other SRAD transmitters, operating in frequency bands other than 440 and 900MHz, should not anticipate frequency assignments. However, ESRA may direct the use of a specific frequency at the time of the Final Flight Safety Review in an effort to assist teams in deconflicting these transmissions.

D&T-4.3.3 With the exception of Primary COTS GPS systems in the 440MHz band, ESRA anticipates making frequency assignments in the 440 and 900MHz, and 1.2, 2.4, 3.3 and 5.8GHz bands, at the time of the Final Flight Safety Review. These assignments will be provided after completing a successful safety review. Since the time available "in the queue" is limited, teams must be able to change transmitter frequencies quickly and without the need to take the rocket apart or remove components from within the rocket. ESRA anticipates that frequency changes will be made using the radio itself, or alternatively, through an externally accessible port or hatch.

D&T-4.3.4 Teams shall not power any transmitting electronics while in the pit area unless authorized by ESRA. This applies to transmitters both within and outside of the COTS frequency ranges within each band (i.e. all COTS and SRAD transmitters). Procedures for obtaining permission to test electronics will be finalized prior to the competition.

- At the present time, it is anticipated that teams will be allowed to test their Primary COTS GPS units in the pit or queue areas after receiving their assigned operating frequencies (see 4.3.2 for specific information on the frequency assignment process for COTS GPS units).
- No other radio transmissions will be allowed until the time of the Final Flight Safety review and as approved by the Queue Manager. Transmitters shall be powered off when not in use
- Low power transmitters, as defined in the SBP, are allowed in the pit area. Transmitters shall be powered off when not in use.

D&T-4.3.5 Teams shall have the ability to quickly change frequencies on their COTS and SRAD transmitting and receiving equipment. If a team is directed to change frequencies, they will not be permitted to fly until such changes are completed and verified.

D&T-4.3.6 Teams shall not transmit on a COTS GPS frequency that is not assigned to them, or on any frequency that is not registered and approved by ESRA. Competition officials may assess penalties for teams that violate this requirement.

[5] Note that ESRA pre-assigned SRAD radio frequencies in 2026 for the 400MHz band. ESRA will attempt to do this in 2027. However, the ability to do this will depend on the number of radios that are expected at the competition.

D&T-4.3.7 Trackers that allow custom names to be assigned to the tracker shall include the Team ID number. If the rocket contains multiple sections (e.g., the rocket and a payload or a two-stage booster and sustainer), the sections should be labeled. For example, Team 999's Section 1 tracker name would be "<T999S1>" (without the quotes). This is to aid in automatic parsing.

D&T-4.3.8 Should an SRAD transmission conflict develop between teams waiting in the queue or at the pads, it shall be the responsibility of the affected teams to resolve the conflict: Options for resolution of a conflict include the following:

- One team changing their tracking system to a different (unused) frequency, or
- One team delaying their launch to a subsequent salvo.

D&T-4.3.9 Transmitter operation associated with ground support equipment for Hybrids and Liquids shall coordinate their operations with the Hybrid Pad Manager.

D&T-4.3.10 Transmitter operation associated with the Live Video Challenge shall be consistent with the SBP and the rules associated with the 2027 Live Video Challenge.

4.4 SRAD GPS Tracking Systems And Other SRAD Transmissions

D&T-4.4.1 SRAD GPS solutions may be used, at the option of the team, subject to the following:

- Transmissions must be outside of the ranges designated for COTS GPS tracking and must meet the requirements listed in the SBP.
- Additional documentation detailing the design, fabrication, operation, and testing of SRAD GPS tracking systems shall be included in progress reports.
- The use of an SRAD GPS tracking system does not remove the requirement to use a Primary COTS GPS transmitter in each independently recovered section of the rocket.

D&T-4.4.2 APRS protocol via 1200 baud AFSK is recommended for SRAD GPS Tracking on the 2m, 70cm and 33cm bands.

D&T-4.4.3 It is recommended to flight-test SRAD GPS systems.

D&T-4.4.4 Neither COTS GPS trackers nor SRAD GPS trackers will be used for official Post Flight Data Recovery altitude determination. Currently, altitude determinations are performed using barometric altitude as outlined in the Rules and Requirements document.

D&T-4.4.5 All SRAD transmissions shall follow the requirements in the SBP

4.5 FCC Amateur Radio Licensing

D&T-4.5.1 Teams are encouraged to get an Amateur Radio Operator's license for their country. However, Teams are permitted to use the operating license of a Mentor, FoR or ESRA Volunteer if such

persons agree in advance to provide this service. In such cases, the control operator must be present any time the radio is in operation.

D&T-4.5.2 70cm and 2m GPS Tracking solutions require that the control operator to be licensed at the Technician level or higher.

D&T-4.5.3 The control operator's call sign shall be used for any transmission requiring an operating license by law or as required in the SBP.

D&T-4.5.4 A HAM License is required for any video transmission from the rocket, in any band, and the HAM Call Sign shall be displayed on the screen.

D&T-4.5.5 The control operator's call sign and team number shall be registered with ESRA via the periodic progress reports.

D&T-4.5.6 Teams utilizing Amateur Radio frequencies and based outside of the US shall secure US operating authority, use a solution transmitting at 900 MHz or another unlicensed band if a reciprocal licensing agreement cannot be obtained or cannot be used, or arrange for an alternative control operator as described in Section 4.5.1. US operating authority is most commonly accomplished via reciprocal licensing agreements.

D&T-4.5.7 International teams using a team-provided operating license shall bring documentation in English (translated if necessary) of their US operating authority if using frequencies outside of an unlicensed band.

4.6 GPS Tracking Solution Safety Inspections

D&T-4.6.1 IREC personnel at the pads shall verify that teams are able to communicate with their receiving station to confirm that their Primary COTS GPS signals are being acquired and are functioning properly.

D&T-4.6.2 Launch operations will not be delayed because of Primary COTS GPS inoperability. Teams which cannot confirm a COTS GPS Tracking signal shall not be allowed to launch.

D&T-4.6.3 The inoperability of an SRAD GPS or any other SRAD transmission will not be a reason to delay launch operations.

4.7 Recovery Team Training and Review at Team Registration

D&T-4.7.1 Prior to arrival Midland on Monday of the week of the IREC, all members of the recovery team shall have completed video training on utilization of the recovery radios or backpack systems.

D&T-4.7.2 Teams may need to verify completion of training during the check-in process in Midland.

D&T-4.7.3 Teams must have a recovery plan before deployment. Teams that do not have good GPS information, or if the location is not known, must develop range and direction estimates in their recovery plan to avoid excessive wandering. The recovery plan shall be reviewed by ESRA personnel before a team will be released to recover their rocket.

5.0 PROPULSION SYSTEMS

5.1 MOTOR CATEGORIES

D&T-5.1.1 COTS Motors

- A Commercial Off the Shelf (COTS) motor is defined as a motor which has been certified by the Canadian Association of Rocketry, Tripoli, or NAR, and appears on the then current combined Certified Rocket Motors List.
- A list of all approved motors can be found on the TRA website:
  - NAR-TMT Combined Motor List - Tripoli Rocketry Association

D&T-5.1.2 SRAD Motors

- Student Research and Developed Motors (SRAD) are defined as:
  - Any motor built in part or whole by students.
  - Any COTS solid motor modified in such a way as to conceivably affect the operation or performance of the motor.[6]
- Commercial solid propellant grains shall not be used in SRAD motors.
- SRAD motors shall be static fired, and the propellant characterized prior to the submission of the Technical Report.
- Static fire test results and propellant characterization data demonstrating a successful motor test shall be provided to the ESRA safety reviewer prior to submitting the Technical Report.
- No second party motors (*i.e.*, not built by the team) shall be permitted.

5.2 PROPULSION TYPES

D&T-5.2.1 Solid Motor

- Any motor utilizing exclusively a solid propellant.
- Solid propellants shall consist of non-hazardous ingredients per Section 5.5.
- SRAD solid motors shall be static fired, well characterized and tested per Section 5.16.
- All SRAD solid motors shall comply with the TRA Unified Safety code.

D&T-5.2.2 Hybrid Motor

- Any motor utilizing a combination of solid and liquid or gaseous propellants.
- Hybrid propellants must be non-hazardous as defined in Section 5.5.
- SRAD hybrid motors shall be static fired, well characterized and tested per Section 5.16.
- SRAD hybrid motors shall comply with the TRA Unified Safety code.
- Nitrous oxide is the only allowable oxidizer.

[6] For example, modification of the forward closure of a motor to support head end ignition would result in the motor being classified as SRAD, as would any modification related to the threads used on the closures. Removal of a thrust ring from the aft closure might not result in a classification as SRAD as this would not affect the operation of the motor.

D&T-5.2.3 Liquid Engine

- An engine utilizing stored fuel and stored oxidizer in the liquid state.
- Liquid engines shall be static fired, well characterized, and tested as per Section 5.16.
- Liquid engines shall comply with the TRA Unified Safety Code.
- Nitrous oxide is the only allowable oxidizer.
- All liquid propellants must be non-hazardous as defined in Section 5.5 and are limited to:
    - Isopropyl alcohol (isopropanol),
    - Ethyl alcohol (ethanol),
    - Methanol, and
    - Mixtures that include other hydrocarbons, dissolved solids, gasoline and/or water, containing at least 70% isopropanol or ethanol by volume, and which are not hypergolic or pyrophoric.

5.3 Rail Departure Velocity

D&T-5.3.1 Teams shall design their rockets to attain a rail departure velocity of at least 25 m/s (82 ft/s) based on the "effective" length of the rail. "Actual" rail lengths for the ESRA launch pads are currently 17 feet for single-stage rockets and 21 feet for two-stage rockets.

D&T-5.3.2 The "effective" rail length is defined as the distance (altitude) that the rocket has traveled at the instant that the launch vehicle first becomes free to move about the pitch, yaw, or roll axis (i.e., the instant when only one rail button is connected to the rail).[7]

D&T-5.3.3 Teams must include the rail length for any team-provided launch pad in their progress reports and then use that rail length in their calculation of the rail departure velocity.

D&T-5.3.4 For tower launchers, the required rail departure velocity must be reached by the time that the center of pressure of the rocket has cleared the tower.

D&T-5.3.5 Cluster rockets shall meet the minimum rail departure velocity requirement on any single motor in the cluster that is ignited on the ground.

5.4 Igniters

D&T-5.4.1 Teams shall supply the appropriate means of ignition for their motor.

D&T-5.4.2 Hybrid/Liquid teams shall discuss their igniter approach with ESRA safety personnel as soon as practicable.

[7] As an example, if a single-stage rocket has two rail buttons located 1foot and 4 feet from the bottom of the rocket, the effective rail length is 17 - 4 = 13 feet. The rail exit velocity is then the velocity of the rocket, as determined from the simulation of flight when the "altitude" of the rocket has reached 13 feet.

D&T-5.4.3 For solid motors:

- Ground-started motors shall be fired with dual-headed or dual-bridged igniters.
  - The use of two high-current igniters should be avoided due to excessive current requirements.
- Motors may be started using either a COTS igniter or an SRAD igniter.
  - Teams shall specify the igniter(s) to be used during the Video Flight Readiness Review.
  - Teams using an SRAD igniter shall specify the design and composition of the igniter during the Video Flight Readiness Review.
  - Teams shall present their igniters for inspection at the Final Flight Safety Review. Any change in the igniter from that presented in the Video Flight Readiness Review must be brought to the attention of the ESRA volunteers at the time of the Final Flight Safety Review.
- COTS igniters that are modified shall be considered SRAD igniters and subject to the requirements of this section.

5.5 NON-HAZARDOUS PROPELLANTS

D&T-5.5.1 Launch vehicles entered in the IREC shall use non-hazardous propellants.

D&T-5.5.2 Ammonium Perchlorate Composite Propellant (APCP), potassium nitrate and sugar (aka rocket candy), nitrous oxide, kerosene, propane, alcohol, and similar substances, are considered non-hazardous.

D&T-5.5.3 Hazardous propellants are defined as those requiring breathing apparatus, unique storage and transport infrastructure, or extensive personal protective equipment (PPE, body suits). Hydrogen peroxide and cryogenic liquids are considered hazardous propellants.

D&T-5.5.4 The use of ammonium nitrate as the oxidizer in solid rocket motors is prohibited.

5.6 SUGAR MOTOR PROPELLANTS

D&T-5.6.1 Sugar propellants shall use potassium nitrate as oxidizer.

D&T-5.6.2 The following sugars or sugar alcohols are allowable as fuel:

- Dextrose,
- Erythritol, and
- Sorbitol.

D&T-5.6.3 Other chemicals may be used if the fuel includes more than 50% by weight of the sugars or sugar alcohols listed in 5.6.2.

5.7 Allowable materials for Case Components for SRAD Motors

D&T-5.7.1 Metallic cases shall be made of non-ferrous ductile metals such as 6061 aluminum alloy.

D&T-5.7.2 Non-metallic cases shall not be made of brittle materials which may rupture into sharp shards, such as PVC or other low-temperature polymers.

D&T-5.7.3 Forward closures shall not be made of ferrous materials.

D&T-5.7.4 Minor components such as snap rings, nozzle washers, rear closures, and seal disks may be made of ferrous metals.

D&T-5.7.5 Nozzles for Sugar Motors may be made of steel as long as the throat of the nozzle is recessed within the case.

D&T-5.7.6 Combustion chambers for hybrid motors shall not be made from any alloy of steel.

5.8 Propulsion System Safing and Arming

D&T-5.8.1 A propulsion system shall be deemed armed if only one action (*e.g.*, an ignition signal) must occur for the propellant(s) to ignite.

D&T-5.8.2 The "arming action" is any action enabling an ignition signal to ignite the propellant(s).

D&T-5.8.3 The IREC-provided launch control system described in Section 11.2 will be used for all solid and two-stage launches.

D&T-5.8.4 Hybrid or liquid teams shall provide launch control systems that ensure their rocket is properly made safe, *i.e.*, not armed until commanded. Additional requirements for team-provided launch control systems are defined in Section 11.3 of this document.

5.9 Propellant Offloading After Launch Abort

D&T-5.9.1 Hybrid and liquid propulsion systems shall implement a means for remotely controlled venting or offloading of all liquid and gaseous propellants.

D&T-5.9.2 All Bi-Liquid and Hybrid Motors with Quick Disconnect (QD) Oxidizer fill plumbing:

- Dump capable valves will be present on both sides (Supply and Flight tanks) of the disconnect. Supply side may be a manual ball valve. The Flight Tank side dumping shall be remotely operated. The Main Oxidizer Flow Valve may be used in the dump role IF it is designed with multiple actuation capability, otherwise a dedicated Flight tank side Dump valve is required.

D&T-5.9.3 Hybrid motors without QD connections:

- There shall be a dump valve somewhere in the fill line system with a direct line access to the Flight tank.

5.10 Ground-Start Ignition Circuit Arming

D&T-5.10.1 All personnel shall be at least 15 m (50 ft) away from the launch vehicle prior to ground-started propulsion system ignition circuits/sequences being armed.

D&T-5.10.2 The IREC provided launch control system satisfies this requirement.

5.11 Air-Start/Staged Ignition Circuit Arming

D&T-5.11.1 All upper-stage (i.e., air-starts) propulsion systems shall be designed to:

Prevent motor ignition during arming on the ground.

Inhibit motor ignition, as defined in Section 5.13, in the event of a non-nominal flight.

Provide the capability of being safely disarmed in the event the rocket is not launched.

5.12 Air-Start/Staged Ignition Circuit Electronics

D&T-5.12.1 All upper-stage ignition systems shall comply with either the same requirements for Safety Critical Wiring as recovery systems as defined in Section 6.12 and Appendix B or be specifically approved by the Two-Stage Pad Manager.

5.13 Air-Start/Staged Flight Computer Requirements

D&T-5.13.1 Ignition of air-start motors shall be accomplished using a COTS flight computer configured to inhibit ignition of the air-start motor(s) unless all the following conditions have been met:

- Booster burnout has been detected.
- The rocket has reached an altitude of at least 70% of the simulated altitude at the time when initiator firing is desired (the simulation shall be performed at a 0° launch angle with no wind).
- The sustainer flight angle is within 20° of vertical.

D&T-5.13.2 The flight computer shall be configured to prevent the air-start motor from firing at a later time if the altitude threshold or flight angle criteria are not achieved.

D&T-5.13.3 Currently available flight computers that satisfy the requirements of this section include, but are not limited to:

- Featherweight Blue Raven and Blue Jay Plus,

- Aim Xtra,
- Multitronix Kate,
- Altus Metrum Telemega and EasyMega, and
- MARSA Systems Marsa33 and Marsa54 flight computers with Tilt Module & Interface.

D&T-5.13.4 SRAD flight computers shall not be used for the purpose of igniting air-start motors.

D&T-5.13.5 Simple timers shall not be used for igniting air-start motors unless in combination with altitude and tilt lockout.

5.14 Air-Start/Staged Flights – Arming Procedures

D&T-5.14.1 Projects shall include provisions that prevent air-start motor ignition on the ground.

D&T-5.14.2 All projects shall include a provision to open the circuit between the flight computer and the air-start motor igniter, and to shunt the igniter, prior to and during power-up of the flight computer.

- This requirements can be accomplished using one or more than one physical switch.
- Examples of recommended switch designs are provided in Appendix C.
- The Multitronix Kate flight computer meets the shunt requirement of D&T-5.14.2.
- The use of SRAD electronics for the purpose of remotely accomplishing the switching functions (isolation and shunt) in this section can be considered; however, under no circumstances can such electronics cause a current to flow to the igniter.

D&T-5.14.3 Flight computers shall not be powered up until the rocket is in a vertical position. This includes flight computers that have an "arming" function as part of their user interface.[8]

- Devices that will power flight computers via remote operation, such as Wi-Fi switches, can be powered up prior to lifting the rocket to a vertical position.

D&T-5.14.4 The electronics configuration shall be designed such that the provisions used to open and to shunt the circuit to the igniter can be used to safe the igniter if the rocket is not launched.

5.15 Air-Start/Staged Flights – Additional Information Requirements

D&T-5.15.1 Teams shall provide additional information in their CDR specifically related to air-start/staged flights.

D&T-5.15.2 Updated information, including a response to ESRA comments as applicable, shall be included in the CDR and discussed during the Video Flight Safety Review.

[8] Teams are encouraged to develop processes that allow the power-up or arming procedures in this section to be conducted from the ground rather than from a ladder.

D&T-5.15.3 The required additional information includes:

- Schematic diagram of the electronics configuration to be used for air-start motor ignition and recovery.
- A graph illustrating flight simulation profile, to include altitude, velocity, and acceleration as a function of time to the expected apogee time.
- An explanation of the strategy for the flight based on the flight profile (i.e., what is the rationale behind the selection of staging times, coast times, etc.).
- A description of the specific procedures that will be used to prevent air-start motor ignition on the ground.
- A description of the specific procedures that will be used to inhibit air-start motor ignition in the event of a non-nominal flight.
- Drawing & description of the interstage coupler.
- Drawing and description of the wiring design between the flight computer and the igniter, including any break wires employed on this wiring.

5.16 SRAD Propulsion System Testing

D&T-5.16.1 Teams shall comply with all rules, regulations, and best practices imposed by the authorities at their chosen test location(s).

D&T-5.16.2 Teams shall complete required testing of SRAD and modified COTS propulsion systems, and submit this information to their ESRA safety reviewer, prior to submitting the Technical Report. Teams unable to make this deadline may switch to COTS but must notify their ESRA safety reviewer prior to submitting their Technical Report.

D&T-5.16.3 Combustion Chamber Pressure Testing:

- SRAD and modified COTS propulsion system combustion chambers shall be designed and tested according to the SRAD pressure vessel requirements defined in Section 6.18 of this document.
- Combustion chambers are exempted from the requirement for a relief device.

D&T-5.16.4 Hybrid and Liquid Propulsion System Tanking Testing:

- SRAD and modified COTS propulsion systems using liquid or hybrid propellant(s) shall successfully complete a propellant loading and off-loading test in “launch-configuration”.
- This test may be conducted using actual propellant(s) or suitable proxy fluids.
- The loading and off-loading test shall demonstrate meeting the fill-to-fire time and detanking time requirements of 5.17.
- Videos and testing data shall be posted on the Team’s shared drive with results discussed in the Technical Report.

D&T-5.16.5 Static Hot-Fire Testing:

- SRAD propulsion systems shall successfully complete an instrumented (chamber pressure and thrust), full scale (including system working time) static hot-fire test prior to the IREC.

- SRAD Solid teams shall provide a BurnSim or openMotor file of their motor.
- SRAD Solid teams shall provide static test data including at least a graph of the pressure and thrust versus time.
- SRAD static test results shall be provided in the Comma Separated Values (CSV) file format and include Time (sec), Pressure (pascals) and Thrust (newtons) data.
- Video of successful testing shall be posted on the Team's shared drive and discussed in the Technical Report.
- The flight motor shall represent the static test motor in all respects.
- Any changes to the SRAD flight motor shall be tested and resubmitted.

5.17 SRAD Hybrid/Liquid Pad Operations Design Requirements

D&T-5.17.1 Fill to fire time shall be ≤30 minutes. The filling time is defined as the time from filling start to launch ignition.

D&T-5.17.2 Pad managers may, using their discretion, permit an additional 15 minutes of fill time.

D&T-5.17.3 Rockets shall vent/detank in ≤30 minutes, including in fault and failsafe conditions.

D&T-5.17.4 Rockets shall be able to hold fully filled for ≥10 minutes before launching.

D&T-5.17.5 Teams will be told to abort if they do not meet these times.

D&T-5.17.6 Launch Pad Oxidizer Quick Disconnect (QD). Quick Disconnect systems shall be designed to be Rocket/Pad position neutral to prevent binding. Swing Arms are allowed IF they can be proven to operate under any conditions of vehicle weight, position or motion while loaded on the rail, "dry" or "wet". All QD systems will be submitted for approval, upon request, by the Hybrid/Liquid pad officials.

6.0 RECOVERY SYSTEMS AND AVIONICS

6.1 DUAL-EVENT PARACHUTE AND PARAFOIL RECOVERY

D&T-6.1.1 Each independently recovered launch vehicle body, including payloads, anticipated to reach an apogee above 457 m (1,500 ft) above ground level (AGL) shall follow a "dual-event" recovery operations concept (CONOPS) defined to include the following:

- A drogue deployment event of a drogue parachute (or a reefed main parachute) at or near apogee.
- A main deployment event of a main parachute, or unreefing of a main parachute, at an altitude no higher than 457 m (1,500 ft).

D&T-6.1.2 Independently recovered bodies (payloads) released at an altitude below 1,500 ft (457 m) AGL, are exempted from dual deployment and may feature only a single/main deployment event, which must occur at or near the point of release.

D&T-6.1.3 Boosters from two-stage rockets are subject to the dual-event provisions of this section.

D&T-6.1.4 A single commercially available release device appropriately rated for large parachutes, *e.g.*, Tinder Rocketry Tender Descender L1, L2, L3, and TD-2, the Rattworks AARD, *etc.*, are permitted subject to the following:

- Dual initiators shall be used.
- Redundant energetics (powder charges) are not required.
- Initiators shall be activated by independent flight computers.
- The device shall be successfully ground tested under a load approximating the load anticipated during flight.
- The Jolly Logic Chute Release (JLCR) is not permitted for any recovery purpose.

D&T-6.1.5 Commercially available line/cable cutters, *e.g.*, Tinder Rocketry Piranha and Mako, Archetype Cable Cutter, *etc.*, are permitted subject to the following:

- Redundant devices, activated by independent flight computers, shall be used.
- The use of dual initiators in a single device does not satisfy 6.1.5.1.

D&T-6.1.6 SRAD release devices and line/cable cutter devices are permitted subject to the following:

- Redundant devices shall be used (for both release and cable cutter devices).
- Activation of redundant devices shall be controlled by independent flight computers.
- The SRAD device shall be successfully ground tested under a load approximating the load anticipated during flight.

D&T-6.1.7 No portion of the rocket or payload can be "guided" during any phase of the recovery process. The term "guided" is intended to include, but is not limited to, the use of GPS or other external sensor input, whether this input comes from on board the rocket or from the ground.

6.2 Recovery Area

D&T-6.2.1 Any hardware drifting outside the safe recovery area or onto properties with restricted access will either be abandoned or recovered at the team's own expense.

D&T-6.2.2 Range Safety personnel may deem a recovery design as unsafe if it is determined that there is a reasonable probability the recovery could depart the safe recovery area.

6.3 Initial Deployment Event

D&T-6.3.1 The initial deployment event shall occur at or near apogee.

D&T-6.3.2 Descent rate calculations shall be based on the launch site elevation.[9]

D&T-6.3.3 The descent rate under drogue shall be between 20-40 m/s (65-131 ft/s) at the point where the main parachute is deployed.

6.4 Main Deployment Event

D&T-6.4.1 The main deployment event for any recovery method shall occur at an altitude no higher than 457 m (1,500 ft) AGL.

D&T-6.4.2 Descent rate calculations shall be based on the launch site elevation.

D&T-6.4.3 The vehicle's descent rate at landing shall be less than 11 m/s (36 ft/s).

6.5 Ejection Gas Protection

D&T-6.5.1 The recovery system shall implement adequate protection (*e.g.*, fire resistant material, pistons, baffles, *etc.*) to prevent ejection energetics from damaging recovery components including, but not limited to, shock cords and parachutes.

6.6 Parachute Swivel Links

D&T-6.6.1 The recovery system rigging (*e.g.*, parachute lines, risers, shock cords, *etc.*) may implement appropriately rated swivel links at connections to relieve twist/torsion as the specific design demands.

[9] The location for the 2027 IREC is not yet known (as of the date this DTEG was published). Teams will use the elevation of the site for their recovery calculations when the location of the new site is determined.

6.7 PARACHUTE COLORATION AND MARKINGS

D&T-6.7.1 When separate parachutes are used for the initial and main deployment events, these parachutes shall be contrasting colors: from each other, blue sky, and clouds.

D&T-6.7.2 Through the use of visually distinct parachute colors and/or patterns, teams should be able to visually determine if a main was deployed at apogee, or was not deployed at the expected deployment altitude during descent (drogue only recovery).

6.8 NON-PARACHUTE/PARAFOIL RECOVERY SYSTEMS

D&T-6.8.1 Teams utilizing non parachute or parafoil-based recovery methods shall notify ESRA of their intentions as early as possible and keep ESRA apprised of their progress.

D&T-6.8.2 ESRA reserves the right to impose additional requests for information and draft unique requirements depending on the team's specific design.

6.9 REDUNDANT ELECTRONICS

D&T-6.9.1 Launch Vehicles shall implement completely independent and redundant recovery systems to include arming switches, sensors/flight computers, power supplies, redundant energetics (except as noted in D&T-6.1.4), and electric initiators.

D&T-6.9.2 A dual pole switch does not satisfy the requirement for redundant arming switches.

6.10 REDUNDANT COTS RECOVERY ELECTRONICS

D&T-6.10.1 At least one of the redundant recovery subsystems in each independently recovered section of the rocket shall implement a COTS flight computer (*e.g.*, StratoLogger, AIM, Kate, EasyMini, TeleMetrum, RRC3, Raven, *etc.*).

D&T-6.10.2 A COTS flight computer shall serve as the official altitude logging system as specified in Section 2.5 of the IREC Rules & Requirements Document.

- This function may be provided by one of the flight computers satisfying 6.9.1 or a separate flight computer.
- A separate flight computer may be armed and powered by the same switch and power supply as one of the flight computers satisfying 6.9.1.

D&T-6.10.3 A COTS flight computer shall fire either the primary or redundant energetic system.

D&T-6.10.4 To be considered COTS, the flight computer (including flight software) shall have been developed and validated by a commercial third party.

D&T-6.10.5 Commercially designed flight computer kits (*e.g.*, the Eggtimer or similar) are not permitted as COTS.

6.11 SRAD Recovery Electronics

D&T-6.11.1 Flight computer kits, such as Eggtimer or similar may be used as SRAD electronics.

D&T-6.11.2 Any SRAD flight computer assembled from separate COTS components shall be considered an SRAD system.

D&T-6.11.3 A COTS microcontroller running SRAD flight software shall be considered an SRAD system.

D&T-6.11.4 A COTS flight controller containing any student modifications to the base software or hardware shall be considered an SRAD system.

D&T-6.11.5 SRAD flight computers shall be documented in the progress reports.

D&T-6.11.6 Teams shall provide proof of function of SRAD flight computers, including ground testing and flight testing before the competition.

- If flight testing cannot be performed, the recovery electronics shall be designed such that a COTS flight computer can be substituted for the SRAD flight computer if requested by the ESRA safety reviewer.
- The ESRA safety reviewer will make a determination on the use of an SRAD flight computer based on the proof of function data provided by the team.

D&T-6.11.7 SRAD recovery electronics shall take appropriate recognition of effects of noise and sensor error. Use of appropriate sensor filtering with testing is required.

6.12 Safety Critical Wiring for Recovery Systems and Air-Start Motors

D&T-6.12.1 Definition:

- Safety critical wiring is any electrical wiring associated with recovery system deployment events, air start motors, location beacons (*e.g.*, GPS), and systems that can affect the rocket trajectory or stability.
- All safety critical wiring shall meet the requirements in this section and the safety critical wiring guidelines described in Appendix B of this document.

D&T-6.12.2 Cable Management:

- All safety critical wiring shall implement a cable management solution (*e.g.*, wire ties, wiring, harnesses, cable raceways) which will prevent tangling and excessive free movement of significant wiring/cable lengths due to expected launch loads.
- Wiring shall be labeled or color-coded to allow easy determination of the subsystem to which it is attached.

- Wiring shall incorporate strain relief at all connections/terminals to prevent unintentional de-mating/disconnection.

D&T-6.12.3 Secure Wiring Connections:

- All safety critical wiring/cable/terminal connections shall be sufficiently secure as to prevent disconnection during handling, transport, launch preparation and flight.
- Electrical connectors shall not be used as structural elements, *e.g.*, the mounting of daughterboards on a mainboard.
- Structural supports (*e.g.*, screws and standoffs), are required for daughterboard mounting.
- The use of twisted wires, even with the use of wire nuts or similar devices, to make or break flight essential connections is prohibited.

6.13 Recovery System Testing

D&T-6.13.1 Teams shall comply with all rules, regulations, and best practices imposed by the authorities at their chosen test location(s).

D&T-6.13.2 The following requirements concern verification testing of all recovery systems.

D&T-6.13.3 A COTS microcontroller running SRAD flight software shall be considered an SRAD system.

D&T-6.13.4 Ground Test Demonstration:

- All recovery system mechanisms shall be successfully tested, without significant anomalies, prior to the IREC through one or more ground tests of key subsystems.
- Sensor electronics shall be functionally included in ground test demonstrations by simulating the environmental conditions under which their deployment functions are triggered (*i.e.*, expected ambient atmospheric conditions).[10]
- Video(s) of ground test demonstrations shall be posted to the Team's shared drive. Note, however, that "pop" testing is allowed at the competition site and proof of successful pop testing, without anomalies, is required prior to flight.

D&T-6.13.5 Flight Test Demonstration:

- A flight test demonstration is recommended and may be required by flight safety personnel in the case of an unusual or untested recovery approach.
- In the case of such a flight test, the recovery system flown shall verify the intended design by implementing the same major subsystem components (*e.g.*, flight computers and parachutes) as will be integrated into the launch vehicle intended for the IREC.
- Video(s) of the test flight(s) shall be posted to the Team's shared drive.

[10] This requirement means that sensor electronics should be ground tested at the conditions they will experience during flight. It does not require that a flight computer itself must be used to trigger a recovery charge during a ground test (i.e., a pop test).

6.14 Stored-Energy Devices – Energetic Device Safing and Arming

D&T-6.14.1 For the purposes of this section, energetics are defined as all stored-energy devices, other than propulsion systems, that have reasonable potential to cause bodily injury upon energy release.

D&T-6.14.2 An energetic device is considered made safe when two separate events are necessary to release the energy.

D&T-6.14.3 An energetic device is considered armed when only one event is necessary to release the energy.

D&T-6.14.4 All energetics shall be made safe until the rocket is in the launch position, at which point they may be armed at the direction of IREC personnel.

D&T-6.14.5 The following table lists some common types of stored-energy devices and overviews in what configuration they are considered non-energetic, made safe, or armed.

Table 6-1 - Common Types of Stored-Energy Devices

DEVICE CLASS	NON-ENERGETIC	MADE SAFE	ARMED
Igniters/Squibs	Small igniters/squibs, nichrome, wire or similar	Large igniters with leads shunted	Large igniters without shunted leads
Pyrogens (*e.g.,* black powder)	Very small quantities contained in non-shrapnel producing devices (*e.g.,* pyro-cutters or pyro-valves)	Large quantities with no igniter, shunted igniter leads, or igniter(s) connected to unpowered avionics	Large quantities with non-shunted igniter or igniter(s) connected to powered avionics
Mechanical Devices (*e.g.,* powerful springs)	De-energized/relaxed state, small devices, or captured devices (*i.e.,* no jettisoned parts)	Mechanically locked and not releasable by a single event	Unlocked and releasable by a single event
Pressure Vessels	Non-charged pressure vessels	Charged vessels with two events required to open main valve	Charged vessels with one event required to open main valve

D&T-6.14.6 Teams shall not bring amounts of pyrogenic materials to the event greater than the amount required for ground testing and the competition flight.

D&T-6.14.7 All stored-energy devices (aka energetics) used in recovery systems shall comply with the energetic device requirements defined in this section.

6.15 ARMING DEVICES

D&T-6.15.1 All energetic device arming features shall be externally actuatable and controllable. Magnetic switches and the Multitronix turn-on coil shall be understood to be externally actuatable and controllable.

D&T-6.15.2 Access panels which may be secured for flight while the vehicle is in the launch position are acceptable.

D&T-6.15.3 All energetic device arming features shall be accessible without any person placing their head against the rocket or within 100 mm of the rocket.

D&T-6.15.4 All energetic device arming features shall be operable while wearing a face shield and any other required PPE.

D&T-6.15.5 All energetic device arming features shall be located on the airframe such that any inadvertent energy release will not impact personnel arming them.[11]

D&T-6.15.6 Teams are responsible for verifying that they will be able to safely reach their arming switches using either ESRA or team-provided infrastructure. Teams that cannot do this will not be allowed to fly. Currently, rockets in the solids area are supported approximately 2 feet above the ground with (at least) 10-foot stepladders available. In the two-stage area, rockets are supported approximately 4 feet above the ground with 24-foot extension ladders available.[12]

- The location of switches requiring manual activation must be included in the OpenRocket file.

6.16 ARMING DEVICE VERIFICATION

D&T-6.16.1 Arming shall be verified without any person placing their head against the rocket or within 100 mm of the rocket. A stethoscope may be used to increase the distance between an audio source and a human ear.

[11] For example, the arming key switch for an energetic device used to deploy a hatch panel shall not be located at the same airframe clocking position as the hatch panel deployed by that charge.

[12] For the ESRA two-stage pads, the ladder support is located approximately 17 feet above the ground (13 feet above the base of the rocket). Teams will not be permitted to climb the ladder to a point where the persons belt buckle is above the ladder support. This corresponds to an eye level position that is about 15 feet above the base of the rocket.

D&T-6.16.2 Arming verification shall be designed to be completed in an environment with at least 76 dBA background noise plus at least 32 km/h (20 mph) wind.

D&T-6.16.3 Arming verification shall be designed to be completed while wearing a face shield and any other required PPE.

6.17 Batteries Contained In The Rocket

D&T-6.17.1 Prohibited Batteries:

- Lithium-Polymer (LiPo) batteries are not permitted due to fire hazard.
- Li-Ion batteries in a rectangular form factor and/or with plastic casing are prohibited. Cylindrical Li-Ion batteries that are arranged in a rectangular container are allowed.
- Wet cell lead acid batteries are prohibited.
- Any nuclear batteries.

D&T-6.17.2 Allowed Batteries:

- LiFePO4 chemistry cells are allowed in any form factor and in any casing (plastic or metallic).
- NiMH (Nickel-Metal hydride) batteries are allowed in metal casing and any form factor.
- Alkaline (non-rechargeable) batteries are allowed in metal casing and any form factor.
- Li-Ion batteries are permitted if packaged in a cylindrical metallic casing.
- Lithium primary coin cell batteries in metal casing are permitted.

D&T-6.17.3 Teams should inquire about battery chemistries and form factors not listed here.

6.18 SRAD Pressure Vessels

D&T-6.18.1 Teams shall comply with all rules, regulations, and best practices imposed by the authorities at their chosen test location(s).

D&T-6.18.2 The following requirements concern design and verification testing of SRAD and modified COTS pressure vessels.

- Unmodified COTS pressure vessels utilized for other than their advertised specifications will be considered modified, and subject to these requirements.

D&T-6.18.3 Relief Device:

- All SRAD pressure vessels shall implement a relief device, set to open at no greater than the proof pressure specified in the following requirements.
- SRAD (including modified COTS) rocket motor propulsion system combustion chambers are exempted from this requirement.

D&T-6.18.4 Designed burst pressure for metallic pressure vessels:

- All SRAD and modified COTS pressure vessels constructed entirely from isotropic materials (*e.g.*, metals) shall be designed to a burst pressure no less than 2 times the maximum expected operating pressure, where the maximum operating pressure is the maximum pressure expected during pre-launch, flight, and recovery operations.

D&T-6.18.5 Designed Burst Pressure for Composite Pressure Vessels:

- All SRAD and modified COTS pressure vessels either constructed entirely from non-isotropic materials (*e.g.*, fiber reinforced plastics; FRP; aka composites), or implementing composite overwrap of a metallic vessel (aka composite overwrapped pressure vessels; COPV), shall be designed to a burst pressure no less than 3 times the maximum expected operating pressure, where the maximum operating pressure is the maximum pressure expected during pre-launch, flight, and recovery operations.

D&T-6.18.6 SRAD Pressure Vessel Testing:

- Proof Pressure Testing – SRAD and modified COTS pressure vessels shall be proof pressure tested successfully to 1.5 times the maximum expected operating pressure for no less than twice the maximum expected system working time. Pressurized water (hydrostatic) testing is recommended unless other ‘as safe' methods are available. The intended flight article(s), that is, the pressure vessel(s), used in proof testing must be the same one(s) flown at the IREC).
- Burst Pressure Testing:
    - There is no requirement for burst pressure testing, however a rigorous verification & validation test plan typically includes a series of both non-destructive (*i.e.,* proof pressure) and destructive (*e.g.*, burst pressure) tests.
    - A series of burst pressure tests performed on the intended design will be viewed favorably; however, this will not be considered an alternative to proof pressure testing of the intended flight article.

7.0 ACTIVE FLIGHT CONTROL SYSTEMS

7.1 RESTRICTED CONTROL FUNCTIONALITY

D&T-7.1.1 All launch vehicle active flight control systems shall be implemented strictly for pitch and/or roll stability augmentation, or for aerodynamic braking.

D&T-7.1.2 Under no circumstances will a launch vehicle entered in the IREC be actively guided towards a designated spatial target or GPS location.

D&T-7.1.3 ESRA reserves the right to make additional requests for information and draft unique requirements depending on the team's specific design.

7.2 UNNECESSARY FOR STABLE FLIGHT

D&T-7.2.1 Launch vehicles implementing active flight controls shall be naturally stable without those controls being implemented (*e.g.*, the launch vehicle may be flown with the control actuator system (CAS) – including any control surfaces – either removed or rendered inert and mechanically neutral, without becoming unstable during ascent).

D&T-7.2.2 Attitude control systems (ACS) shall only mitigate small perturbations which affect the trajectory of a stable rocket that implements only fixed aerodynamic surfaces for stability. However, the use of an ACS to achieve a vertical flight trajectory is acceptable.

D&T-7.2.3 Stability is defined in Section 10.2 of this document.

D&T-7.2.4 ESRA reserves the right to make additional requests for information and draft unique requirements depending on the team's specific design.

7.3 DESIGNED TO FAIL SAFE

D&T-7.3.1 Control actuator systems (CAS) shall default to a neutral state whenever either an abort signal is received for any reason or primary system power is lost.

D&T-7.3.2 A neutral state is defined as one which does not apply any moments to the launch vehicle (*e.g.*, aerodynamic surfaces trimmed or retracted, gas jets off, *etc.*).

7.4 ACTIVE FLIGHT CONTROL SYSTEM ELECTRONICS

D&T-7.4.1 All active control systems shall comply with the "safety critical wiring" requirements, 6.12.

D&T-7.4.2 Flight control systems are exempt from the requirement for COTS redundancy, given that such components are generally unavailable as COTS to the amateur high-power rocketry community.

7.5 ACTIVE FLIGHT CONTROL SYSTEM ENERGETICS

D&T-7.5.1 All stored-energy devices used in an active flight control system (aka energetics) shall comply with the energetic device requirements defined in Section 6.14 of this document.

8.0 AIRFRAME STRUCTURES

8.1 ADEQUATE VENTING

D&T-8.1.1 Launch vehicles shall be adequately vented to prevent internal pressures developing during flight and causing either damage to the airframe or any other unplanned configuration changes.

D&T-8.1.2 Teams shall discuss airframe and altimeter bay venting as part of their Video Flight Readiness Review.

8.2 OVERALL STRUCTURAL INTEGRITY

D&T-8.2.1 Launch vehicles shall be constructed to withstand the operating stresses and retain structural integrity under the conditions encountered during handling and transportation and during rocket flight.

D&T-8.2.2 Teams shall calculate the fin flutter velocity of the rocket and present the results of the calculations in progress reports.

D&T-8.2.3 Teams shall provide a value for the shear modulus of the fin material if that value is used in the calculation of the fin flutter velocity. Values may be estimated, based on supplier information, or determined from actual measurements.

8.3 MATERIAL PROHIBITIONS

D&T-8.3.1 Rockets shall be built using lightweight materials, *e.g.*, fiberglass and carbon fiber, or when necessary ductile lightweight metals (*e.g.*, aluminum), and construction techniques suitable for the planned flight. Teams should refer to the Tripoli Unified Safety Code, and specifically, to "additional information" presented on the Tripoli Website regarding the use of metal in rocket construction.[13]

D&T-8.3.2 PVC (and similar low-temperature polymers), Public Missiles Ltd. (PML) Quantum Tube, and steel (including stainless-steel) shall not be used in any rocket airframe structure. This includes fins, body tubes, and nosecones.

D&T-8.3.3 PVC (and similar low-temperature polymers), stainless steel, or other frangible materials shall not be used for a solid or hybrid propulsion system combustion chamber.

[13] Tripoli is currently considering an update to the Unified Safety Code that may clarify the use of metals in rockets. For the time being, however, the use of significant quantities of metal in rockets is discouraged. Steel, along with other heavy or toxic metals, will not be allowed for airframe structures or for ballast. Steel may be used for components such as recovery system attachment points, minor components in motors, and for sugar motor nozzles as described in the USC. Titanium, including 3D printed titanium and other metals, is not allowed for fins, nose cone tips or other such uses. Note that the use of steel cubesat frames is specifically prohibited.

D&T-8.3.4 Environmentally hazardous materials shall not be used as ballast. This includes lead, mercury, and uranium.

8.4 Load Bearing Eye Bolts And U-Bolts

D&T-8.4.1 All load bearing eye bolts shall be of the closed-eye, forged type.

D&T-8.4.2 All load bearing eye bolts, U-bolts, and links shall be steel.

D&T-8.4.3 This requirement extends to any bolt and eye-nut assembly used in place of an eyebolt.

D&T-8.4.4 Stainless steel components (eye bolts, U-bolts, links, *etc.*) are permissible for use in recovery systems.

8.5 Joints Implementing Coupling Tubes

D&T-8.5.1 Airframe-to-coupler sliding joints intended to separate during a recovery event:

- Joints shall be designed such that the coupling tube extends no less than 1 body tube diameter (1 caliber) into the airframe section from which the coupler will separate during flight.
- A sliding joint of less than one caliber will be allowed between a COTS nose cone and the nose cone coupler to allow the so-called Head-End Dual Deploy (HEDD) approach. The overlap must exceed at least one-half of the body tube diameter (1/2 caliber) and the coupler tube must be inserted to the maximum depth allowed by the design of the nose cone.

D&T-8.5.2 Joints not intended to separate during flight:

- Joints shall be designed such that the coupling tube extends into the mating component to the lesser of 1 body tube diameter (1 caliber) or the maximum depth possible by the design of the mating components.
- Joints shall be affixed by mechanical fasteners and/or permanent adhesive.

D&T-8.5.3 Regardless of implementation (*e.g.*, RADAX or other joint types) airframe joints shall prevent bending, see https://www.osti.gov/biblio/5007820.

8.6 Rail Buttons

D&T-8.6.1 Rail buttons shall implement “hard points” for sliding mechanical attachment of the rocket to the IREC supplied 1515 launch rail, serving to guide the rocket during the initial phase of boost until the rocket achieves sufficient velocity for the fins to provide aerodynamic stabilization.

D&T-8.6.2 A minimum of two (2) rail buttons shall be used, and the locations of the rail buttons shall be clearly indicated in the OpenRocket simulation required in progress reports.

D&T-8.6.3 Airframe attachment points for rail buttons shall be reinforced.

D&T-8.6.4 Rail buttons shall be attached using at least one metallic fastener through the reinforced airframe.

D&T-8.6.5 Adhesive-only attachment is not permitted.

D&T-8.6.6 Fly-away rail guides are not permitted.

D&T-8.6.7 Rail buttons 3D printed in a polymer material are not permitted.

D&T-8.6.8 Conformal, adhesively attached rail guides of a linear shape sliding in the slot in the launch rail are not permitted.

D&T-8.6.9 The rail buttons shall be capable of supporting the launch vehicle's fully loaded weight while the rocket is horizontal and hung from a rail (underslung). The attachment procedure and strength of the attachment will be evaluated during the Horseshoe Pavillion Flight Safety Review.

D&T-8.6.10 Rail button placement shall not result in the rail blocking access to arming electronics and should also not interfere with disassembling the rocket.

D&T-8.6.11 Compliance with the above requirements may not be possible for minimum diameter rockets. In such cases, the rail button attachment approach shall be documented in progress reports and is subject to ESRA approval.

D&T-8.6.12 Teams utilizing launch towers shall provide the launch tower and are exempt from the Rail Button requirements.

8.7 Shear Pins

D&T-8.7.1 Teams shall indicate the locations of shear pins on drawings or simulations of their rocket.

D&T-8.7.2 Teams shall describe the number and size of shear pins used in various sections of the rocket, and how this was determined, in their recovery system discussion in the CDR.

8.8 Identifying Markings

D&T-8.8.1 All rockets shall be labeled with the school name, the Team ID number, and the assigned GPS tracking frequency.

- This label shall be duplicated on each part of the rocket which could separate either as designed or accidentally.
- There are no specific size or format requirements for these labels.

D&T-8.8.2 The booster airframe shall prominently display the Team ID number forward of each fin. The digits of the Team ID shall be marked in sequence, longitudinally, from fore to aft along the booster airframe. See Appendix E for an example.

- Individual digits shall be oriented with their vertical axis in-line with the longitudinal axis of the rocket and shall be readable with the rocket in a vertical position.
- The Team ID digits shall each be at least 75 mm (3 in) high, at least 12 mm (½ in) stroke width, and a solid color that clearly contrasts with a solid background color.
- The Team ID number should be clear to livestream viewers when the rocket is on the pad.

8.9 OTHER MARKINGS

D&T-8.9.1 There are no requirements for airframe coloration or markings beyond those specified in Section 8.8 of this document; however, ESRA offers the following recommendations to student teams.

- Mostly white or lighter tinted color (e.g., yellow, red, orange, etc.) airframes are especially conducive to mitigating some of the solar heating experienced in the IREC launch environment.
- High-visibility schemes (e.g., high-contrast black, orange, red, etc.) and roll patterns (e.g., contrasting stripes, etc.) may allow ground-based observers to more easily track and record the vehicle's trajectory with high-power optics.
- Any form of green or brown, or other colors associated with camouflage patterns, are highly discouraged.

9.0 PAYLOAD

9.1 PAYLOAD RECOVERY

D&T-9.1.1 Payloads may be deployable independent of the launch vehicle, remain inside the launch vehicle airframe, be deployed and remain attached to the launch vehicle throughout the flight, or be deployed on the ground

D&T-9.1.2 Deployable payloads not remaining attached to the launch vehicle airframe, where separation is done prior to landing, shall:

- Incorporate an independent recovery system, reducing the payload's descent velocity to less than 11 m/s (36 ft/s) at touchdown.
- Contains a COTS GPS tracking system that complies with Section 4.2.

D&T-9.1.3 Deployable payloads may not implement gliders, drones, balloons, or any other UAS system.

D&T-9.1.4 Rockets with deployed payloads shall contain no more than a single (1) independently recoverable body.

NOTE: Payloads that are deployed after landing are not considered deployable payloads for the purposes of this section.

9.2 PAYLOAD RECOVERY SYSTEMS

D&T-9.2.1 Payloads shall implement recovery systems that comply with all the requirements outlined in Sections 6.1 through 6.17. Mentions of "rocket" shall be understood to be the payload.

D&T-9.2.2 Payload batteries shall conform to requirements in Section 6.17.

9.3 PAYLOAD ENERGETIC DEVICES

D&T-9.2.1 Payload energetics shall be limited to the Payload Recovery Systems (Section 9.2). All other forms and uses of pyrotechnics and other energetics are not permitted.

9.4 PAYLOAD IDENTIFYING MARKINGS

D&T-9.4.1 All payloads shall be labeled with the school name, Team ID number, and the assigned GPS tracking frequency.[14]

[14] In some cases, frequencies will not be assigned until the time of the Final Flight Safety Review, and the requirement to list the assigned GPS tracking frequency on the payload is waived.

D&T-9.4.2 This label shall be duplicated on each part of the payload which could separate either as designed or accidentally.

9.5 PAYLOAD LOCATION

D&T-9.5.1 The payload on two-stage rockets shall be in the sustainer.

10.0 LAUNCH AND ASCENT TRAJECTORY REQUIREMENTS

10.1 LAUNCH AZIMUTH AND ELEVATION

D&T-10.1.1 Launch officials will set the launch azimuth elevation. Single-stage rockets are typically launched at an elevation angle of 84° ±2° and multi-stage flights are typically launched at 87° ±2°.

D&T-10.1.2 Range Safety Officers reserve the right to require certain vehicles' launch elevation be lower or higher if flight safety issues are identified during pre-launch activities.

10.2 ASCENT STABILITY

D&T-10.2.1 Launch vehicles shall maintain a specified minimum stability margin through the flight, regardless of Cg movement and/or shifting center of pressure Cp location, from launch through motor burnout.

- For subsonic flights, the minimum stability margin is 7.5% of the length of the rocket.
- For supersonic flights (i.e., supersonic at any point in the flight), the minimum stability margin is 10% of the length of the rocket.

D&T-10.2.2 For two-stage rockets, the above minimum stability margins shall be determined separately for the stack and for the sustainer.

D&T-10.2.3 Teams shall provide simulation results showing the stability margin in calibers through the entire flight where the simulation is performed with no wind. The initial stability, in percent, shall be provided.[15]

D&T-10.2.4 Regardless of the above stability results, Teams shall not design rockets with a fin span less than 0.8 calibers, based on the largest diameter tube in the rocket (or in the stage for two-stage rockets).[16]

10.3 OVER-STABILITY

D&T-10.3.1 Launch vehicles shall not exceed a specified maximum stability margin.

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[15] The stability margin should be determined from changes in the center of gravity through the flight combined with changes in the center of pressure as typically provided from rocket simulation programs such as OpenRocket. The specified methodology for determining rocket stability is intended to provide IREC reviewers with a consistent approach for evaluating the basic stability of the rocket (i.e., that the rocket is likely to go "up" when launched). It is not intended to replace a more detailed evaluation of aerodynamic characteristics of the rocket that a team may wish to conduct.

[16] The intent of the fin span rule is to discourage the use of unreasonably small fins, even if stability calculations from a simulation program show that the rocket is stable. Small fin spans are particularly discouraged if the fin is "shadowed" by a larger-diameter tube above the fins.

- The stability margin at launch shall not exceed 18% of the length of the rocket.
- The stability margin through the flight shall not exceed 25% of the length of the rocket.

D&T-10.3.2 Over-stability will be evaluated based on the simulation results described in 10.2.3.

11.0 ESRA Provided Launch Support Equipment

11.1 ESRA-provided Launch Rails

D&T-11.1.1 All teams competing in the solids (COTS or SRAD) and two-stage categories shall use IREC supplied launch control systems.

D&T-11.1.2 ESRA shall provide launch rails for single-stage "solids" rockets measuring at least 5.2 m (17 ft) long, 1.5" x 1.5" (aka 1515) aluminum guide rails of the 80/20® type, see https://8020.net/.

D&T-11.1.3 ESRA shall provide launch rails for two-stage rockets measuring at least 6.4 m (21 ft) long.

D&T-11.1.4 Teams shall contact the Hybrid Pad Manager for detailed information on the rails available for hybrid and liquid rockets.

D&T-11.1.5 Rockets shall be loaded horizontally from the top of the guide rail with the rail then erected to the required launch elevation.

D&T-11.1.6 Once erected, the launch vehicle shall be supported vertically by a submerged mechanical stop (aka "standoff") in the rail.

- The position of the mechanical stop may be adjusted.
- Teams can provide their own standoff. Team-provided standoffs will be subject to safety review, must be clearly marked with the Team ID number, and must be removable with commonly available tools.[17]

D&T-11.1.7 Teams whose designs anticipate requiring a longer launch rail to achieve stability during launch shall provide their own launch pad and rail.

11.2 ESRA-Provided Launch Control System

D&T-11.2.1 The IREC will utilize a Wilson F/X Wireless Launch Control System consisting of one LCU-64x launch control unit and PBU-8w encrypted wireless pad relay boxes.

D&T-11.2.2 Solids and two-stage rockets must use the ESRA-provided launch control equipment.

[17] Standoffs may be designed using the types of fittings commonly used for 1515 rails connected to the front face of the rail (i.e., the face that holds the rail buttons). Team-provided standoffs are allowed primarily for the purpose of helping to avoid damage to the base or nozzle of the rocket and to avoid unsafe methods of supporting the rocket (e.g., wooden blocks).

11.3 TEAM-PROVIDED LAUNCH SUPPORT EQUIPMENT

D&T-11.3.1 Teams are allowed to provide their own launch pads or towers. However, teams are required to provide a justification for why they are not able to use ESRA-provided equipment. The design of this equipment shall be documented in progress reports.

D&T-11.3.2 Team-provided launch pads or towers shall be subject to safety review by ESRA personnel prior to use. Pads or towers deemed unsafe will not be permitted. Teams in this situation will be allowed to use ESRA-provided launch pads.

D&T-11.3.3 Hybrid and Liquid Launch Controllers:

- Hybrid and Liquid Launch controllers shall include remote instrumentation for monitoring oxidizer fill level. This instrumentation will function regardless of the potentially low relative humidity of the launch site (<5%) as instrumentation relying on condensation of water in the air, is unlikely to function. Suggestions are Mass load (load cells), Capacitance, Temperature shifts, etc. Time is NOT an option.
- Hybrid and Liquid Launch controllers are required to work remotely over 2000'(610 m) with an Ethernet (RJ45 style) connection through ESRA's provided Wi-Fi network or a team backup coordinated with the Hybrid Pad Manager. All team-provided wireless launch control systems, if needed and approved, shall be able to displace their antennas at least 30 meters in any direction from the launch point and elevate any antennas to provide line of sight clearance above bushes, terrain, vehicles and personnel.
- Hybrid and Liquid Launch controllers shall have a manually selected igniter switch/button of a momentary (Press On-Release off) type that when released, immediately aborts ALL igniter functions.
- Hybrid and Liquid Launch controllers may have automated functions such as all aspects monitoring and aborting (or inhibiting), but they shall not have automated countdowns nor automated (hands off) ignition sequences. They may have automated scheduling of valves, heaters, etc. (Pre-ignition) but shall not initiate igniters. Control of this Preignition shall also be tied to a momentary (Press On) switch/button that when released, immediately aborts ALL Pre-Ignition functions.
- Hybrid and Liquid Launch Teams shall provide a launch checklist to the Hybrid Pad Manager by the Critical Design Review Deadline (Currently 1 Mar 2027). This Checklist will only include the 'launch' period from the fallback to the safety zone to the ignition/flight of the motor and any anomaly (abort, failure, misfire) appendices. This submission shall be, at a minimum, on the shared drive and accessible by the Hybrid Pad Manager. It may be edited at any time after submission, but the final (hard) copy shall be brought to the Hybrid Flight Readiness Review Table in Midland.

D&T-11.3.4 Fault tolerance and arming

- All team-provided launch control systems shall be at least single fault tolerant by implementing a removable safety interlock (i.e., a jumper or key to be kept in possession of the arming crew during arming) in series with the launch switch.
- Hybrid/Liquid launch controllers shall include a failsafe mechanism providing automatic venting of oxidizer in the event of power or communication loss during fill.

D&T-11.3.5 Ground Power:

- Ground power umbilicals are permitted.
- Ground power equipment shall not be connected to any ESRA equipment.

- Generators are prohibited. Hybrid teams that require the usage of generators may be exempt from this requirement after coordinating and obtaining explicit approval from the Hybrid Pad Manager on a case-by-case basis
- Excessive noise from the ground power support equipment (affecting launch operations) is prohibited.
- Teams shall collect ground power equipment as soon as practical after use (i.e., the next time the pad bank opens for access).
- Team ground power equipment shall be labeled with team number and contact information.
- Wet cell lead acid batteries are prohibited.
- Teams anticipating the use of Li-Po batteries for ground power shall provide details in their progress reports and notify their respective ESRA pad managers.

APPENDIX A — ACRONYMS, ABBREVIATIONS AND TERMS

ACRONYM	DEFINITION
ACS	Attitude Control System
AGL	Above Ground Level
APCP	Ammonium Perchlorate Composite Propellant
AWG	American Wire Gauge
CAS	Control Actuator System
CFR	Code of Federal Regulations
Cg	Center of Gravity
CONOPS	Concept of Operations
COPV	Composite Overwrapped Pressure Vessel
COTS	Commercial Off-the-Shelf
Cp	Center of Pressure
EIRP	Equivalent Isotropic Radiated Power
ESRA	Experimental Sounding Rocket Association
FoR	Flyer of Record
FRP	Fiber Reinforced Plastic
IREC	International Rocket Engineering Competition
LOX	Liquid Oxygen
MCOTS	Modified Commercial Off The Shelf
MSL	Mean Sea Level
NFPA	National Fire Protection Association
SBP	Student Band Plan
SRAD	Student Researched & Developed
TBD	To Be Determined
TBR	To Be Resolved

ACRONYM	DEFINITION
TRA	Tripoli Rocketry Association

TERM	DEFINITION
Amateur Rocket	14 CFR, Part 1, 1.1 defines an amateur rocket as an unmanned rocket that is “propelled by a motor, or motors having a combined total impulse of 889,600 Newton-seconds (200,000 pound-seconds) or less, and cannot reach an altitude greater than 150 kilometers (93.2 statute miles) above the earth’s surface”.
Body Caliber	A unit of measure equivalent to the diameter of the launch vehicle airframe in question.
Excessive Damage	Excessive damage is defined as any damage to the point that, if the systems intended consumables were replenished, it could not be launched again safely. Intended Consumables refers to those items which are – within reason – expected to be serviced/replaced following a nominal mission (*e.g.,* propellants, pressurizing gases, energetic devices), and may be extended to include replacement of damaged fins or boat tails, specifically designed for easy, rapid replacement if such components are on hand and can reasonably be replaced within 30 minutes.
FAA Class 2 Amateur Rocket	14 CFR, Part 101, Subpart C, 101.22 defines a Class 2 Amateur Rocket (aka High Power Rocket) as “an amateur rocket other than a model rocket that is propelled by a motor or motors having a combined total impulse of 40,960 Newton-seconds (9,208 pound-seconds) or less.”
Non-hazardous Propellants	For the purposes of the IREC, the event organizers consider ammonium perchlorate composite propellant (APCP), potassium nitrate and sugar (aka “rocket candy”), nitrous oxide, ethanol, alcohol, methanol and similar, as non-hazardous propellants. Hazardous propellants are defined as those substances requiring breathing apparatus, special storage and transport infrastructure, or extensive personal protective equipment, (body suits, *etc.).* Hydrogen peroxide and cryogenic liquids are considered forms of hazardous propellants.[18]
TRA Adult Flier	A TRA-insured flier who is 18 years old or older

[18] As indicated in the table, hydrogen peroxide is considered a hazardous propellant. The current Rules & Requirements document indicates hydrogen peroxide as non-hazardous. This DTEG designation for hydrogen peroxide and other hazardous propellants will be used for the 2027 IREC.

APPENDIX B — SAFETY CRITICAL AVIONICS WIRING GUIDELINES

B1.0 Introduction

- This appendix is intended to set out requirements for safety critical avionics for all phases of the rocket's operations, including storage, transportation, flight preparation, arming/safing, flight, recovery, and post flight.
- This appendix is intended to set out guidelines for the manufacture and assembly of safety critical flight avionics.
- Safety critical avionics are those associated with recovery system deployments, air-start rocket motors, location beacons (e.g., GPS), and any systems that can affect the rocket trajectory or stability.
- It is recommended that the requirements and guidelines identified herein be applied to avionics that are not associated with safety critical functions to improve the reliability and safety of the non-safety critical systems.
- MIL-HDBK-454, Department of Defense Handbook, General Guidelines for Electronic Equipment, is an excellent reference document for avionics design and construction. It is downloadable from the internet.
- The DTEG is the superseding document in the event of a conflict between the DTEG and MIL-HDBK-454.

B2.0 Hazard Identification

- Any electronic system that utilizes voltages 50 volts (AC or DC) or higher shall be identified with the following warning external to that electronics enclosure; “CAUTION: High voltage inside. Internal access by team personnel only.” Warnings shall be in a typeface at least 0.375 inches tall in a minimum of two locations.
- Teams shall provide the rationale, in their progress reports, for the use of voltages exceeding 24 volts.
- Teams shall identify any unique safety procedures required during arming and safing of high voltage systems.

B3.0 Requirements (Compliance is Required)

B3.1 Wire

- All point-to-point wiring shall be stranded.
- Copper conductors shall be used. Conductors plated with silver or tin may be used.
- Solid wire shall only allowable when it is an integral part of an off-the-shelf component (e.g., E-match leads), fully supported as a cut and jumper on a circuit board, or used for wire wrap connections.
- Wire gauge shall be appropriate for the anticipated current loads. Typical current ratings for AWG wires are shown in the table below.
- Wire gauge shall be appropriate for the expected wire length to minimize voltage loss. The typical resistance for a conductor is shown below.

Table 11-1 - Wire Gauge To Standard Resistance Expected

AMERICAN WIRE GAGE (AWG)	OPERATING CURRENT (AMPS)	RESISTANCE (MILLIOHMS/METER)
18	1.65	23
20	1.04	36
22	0.65	57
24	0.41	91
26	0.26	145
28	0.16	231
30	0.10	368

The current table is conservative, with the recommendations appropriate for continuous use. But the current, resistance and voltage drop for any wire needs to be evaluated.

- Wiring shall be color coded and/or labeled to allow easy traceability to the system it supports (e.g., primary ejection avionics, secondary ejection avionics, GPS locator).

NOTE: The smaller the number for the American Wire Gauge (AWG) the larger the wire and conductor diameter the greater the wire current capability, and the lower the wire resistance.

B3.2 Terminations

- Any wire-twisting splice method (including wire nuts) is explicitly forbidden.
- E-match or similar devices shall be terminated by one of the methods described in B4.1.
- Wago connectors shall be solidly mounted to a surface (preferably horizontal on a bulkhead); Wago connectors shall not be permitted to "hang free".
- All wire terminations shall be inspected before flight.
- All terminations will be examined for exposed conductors that may have a potential for electrical shorts.
- All terminations shall be pull tested by exerting a moderate pull on the wiring to verify no looseness in conductors and that the conductor remains attached to the terminal.
- Soldered connection will be inspected for adequate solder application, cracking, and no "cold" solder joints.
- Acid core flux (typically used in plumber's applications) shall not be used for electrical connections.
- Terminations screwed into a binding screw terminal:
- When a bare conductor is held down by a binding screw terminal the wire shall make a 180-degree hook, and strands shall be visible exiting the screw head. Only one wire shall be permitted per screw.
- The wire bend shall be clockwise, so that it will tighten as the screw is torqued.
- When ring or spade terminals are held down by binding screw terminals, a maximum of two terminals shall be allowed per screw.
- Cylindrical or metal-cased prismatic batteries (in allowable chemistries) requiring soldered connections shall have a means of soldering to the battery (e.g., spot-welded solder tabs.
- Soldered connections directly to cylindrical or metal-cased prismatic batteries shall not be permitted.

NOTE: Exposed conductors are often noted during inspections at termination blocks on circuit boards.

B3.3 Connector/Harness Requirements

- No adhesives (e.g., RTV), or coatings (e.g., Glyptal insulating paint), shall be used to insulate or bundle any element of a wire harness.
- Tapes of any kind (e.g., masking, duct or duck, vinyl electrical, Kapton), shall not be used for insulation of wire terminations (if applicable) or splices.
    - Masking tape is allowable to create tags to identify wire connections
    - Wire terminations and splices shall be insulated by heat shrink tubing as applicable
- Connectors shall be identified by labels and/or color coding to allow determination of the associated system and mating parts.
- Mated connections shall have some device (e.g., connector locks, lacing, or tie wraps), to prevent accidental disconnects.
- Interconnect wiring shall be neat and orderly.
- Payload wiring and connectors shall not be intermingled with flight safety critical wiring and harnesses.
- Harnesses shall be supported by plastic P-clamps, tie bases, or by tie wraps to structure.
- Harness routing shall not be contact sharp edges (e.g., screw threads), that may damage wiring. All wiring shall have strain relief (a small amount of slack) to prevent terminations and connections from demating or intermittent connections.
- Wiring shall be cut to the required length with allowances for strain relief and "service loops" to mate/demate major assemblies.
- Excess wiring shall not be simply bundled within the electronics bay.

B3.4 Component Installation

- The mounting orientation shall minimize movement and prevent accidental operation (e.g., of switches), during acceleration.
- The mounting orientation shall avoid conditions where spring loaded contact may be lost.
- All interconnected items (e.g., barrier blocks, sensors, batteries, actuators, switches, etc.), shall be rigidly fixed to the rocket structure so that they cannot move.
- Switches shall be of the type that are not easily operated by handling and/or inertial forces.

B3.5 Batteries

- Battery connections shall be clearly identified for polarity.
- Batteries shall be mounted and connected appropriately:
- "Velcro" (or similar hook and loop material) and/or tie wraps are not permitted as the sole method of mounting batteries.
- Tapes of any kind (e.g., masking, duct or duck, vinyl electrical, Kapton), shall not be used as a mounting method for batteries.
- Batteries vulnerable to puncture damage (if permitted) shall be mounted in a fashion and location to mitigate the likelihood of such damage during a hard landing or other high impact event.
- Batteries shall be mounted in a direction and manner that prevents acceleration forces from disconnecting the battery connector.

B3.6 Circuit Board Guidelines

- Solderless breadboards (also known as plug-in breadboards) shall not be used.
- Circuit boards using screws shall be mounted on standoffs with clearance between the bottom of the board and the mounting structure.
- An adequate number of mounting locations shall be used to avoid stressing the circuit board.
- Alternate methods of mounting circuit boards, e.g. slide in mounting or card cages, shall adequately support the board to prevent flexing or coming free from flight forces.
- All components greater than 7g shall be staked to prevent motion under acceleration.
- All socketed components (e.g., integrated circuits), and press-fit contacts, shall be positively restrained so that they cannot come loose under vibration.
- The connector between the motherboard and daughterboards shall not be the sole means of support.
- Mechanical supports (e.g., fasteners and standoffs), for the daughterboard are required.
- Cuts and jumpers are permitted if the jumpers are fully staked to prevent damage.

B4.0 Guidelines for Manufacture and Assembly

- This section provides guidance for the manufacture and assembly of avionics.

B4.1 Terminations

- Wire should be stripped only with a wire stripping tool of the correct gauge.
- Severed strands are cause for rejection.
- Removal of wire strands to allow a wire to fit into a smaller hole or terminal is prohibited. Use a smaller diameter conductor or a different terminal type.
- Each end of a wire should be terminated in one of the following approved methods:
- Crimped into a crimp terminal (preferred). This includes crimp terminals on multi-conductor connectors (e.g., 9-pin D-sub connectors).
- All crimp operations should be performed with the correct tooling, using crimp terminals sized for the appropriate wire gauge.
- Where multiple wires are crimped into a single terminal, calculate the effective gauge (for example, two 22 AWG are effectively 19 AWG), and use the appropriately sized terminal.
- Insulated plastic sleeves on terminals are typically not required if they are mounted properly in barrier blocks.
- If insulation is needed, use clear heat-shrink tubing.
- Crimp terminals with insulated plastic sleeves (usually color-coded to indicate barrel size) should be avoided because of the difficulty of inspection. If a terminal is supplied with an insulated plastic sleeve it may be removed and clear heat shrink applied instead.
- Screwed into a binding screw terminal.
- When a bare conductor is held down by a binding screw terminal the wire will make a 180-degree hook, and strands will be visible exiting the screw head. Only one wire is permitted per screw.
- The wire bend will be clockwise, so that it will tighten as the screw is torqued.
- When ring or spade terminals are held down by binding screw terminals, a maximum of two terminals are allowed per screw.
    - Terminated into a circuit board mounted terminal block if the off-the-shelf equipment (e.g., recovery electronics), has built-in terminal blocks.
    - Two-piece terminal blocks shall be positively secured together – friction fit is insufficient.
    - Wires will be terminated by soldering only if a piece of off-the-shelf equipment (e.g., an arming switch), has built-in solder terminals.
- If two or more wires must be joined, one of the following approved methods should be used:
- A maximum of three wires will be crimped into a single terminal barrel.
- Butt-splice terminals are considered to have separate barrels on each end.
- Crimp a ring terminal onto each wire and then screw them into a barrier block. Add approved barrier block jumper pieces if many wires must be joined.
- Screw the bare wire under a binding head screw in a barrier block. Add approved barrier block jumper pieces if many wires must be joined.
- Crimp the wires into an uninsulated butt-splice terminal and then insulate with clear heat-shrink tubing.
- Wago lever style connectors are permitted.
- Wago connectors must be mounted to a surface, preferably a horizonal surface like a bulkhead.
- All insulating tubing (usually heat-shrink) should be transparent. This allows inspection of the underlying hardware.

B4.2 Connector/Wire Harness Guidelines

- No adhesives or coatings (e.g., Glyptal insulating paint), should be be used to insulate or bundle any element of a wire harness.
- Connectors should be identified by labels and/or color-coded to allow determination of the associated system and mating parts.
- Connectors (except coaxial connectors) will use crimp contacts unless no alternative can be procured.
- Coaxial connectors may be soldered.
- Mated connectors (except threaded coaxial connectors) should utilize a positive locking mechanism to keep the two halves mated under vibration and tension.
- Connector mate/demate forces should be considered for circuit board mounted connectors to avaoid stressing solder joints and circuit boards.
- If the mated connectors do not have a positive locking mechanism built-in, they shall utilize an external means of preventing connection separation (e.g., tie wraps, lacing, mechanical clamps).
- Threaded coaxial connectors will be checked for finger tightness.
- Power connectors should be polarized to prevent reversed power application.
- Good practice recommends that individual wires be bundled together to make a harness (factory multi-conductor wiring in a common outer jacket is acceptable).
- It is advisable to separate wiring carrying signals from power wiring to avoid interference issues especially if the power wiring is going to a “noisy” device (e.g., a brushed motor).
- Payload wiring and connectors should not be intermingled with flight safety critical wiring and harnesses.
- Bundling may be accomplished by:
- A light twist (for mechanical reasons only, no EMC mitigation is intended).
- Short (1 cm) lengths of clear heat-shrink tubing, zip-ties, or lacing every 5 cm.
- Plastic “Expando” mesh or split loom plastic sleeving over the wiring harness
- Harnesses may be supported by plastic P-clamps, tie bases, or by tie wraps to structure.
- Certain insulating materials exhibit a cold flow characteristic. Caution should be used in the selection of these materials in applications requiring restrictive clamping or tying, etc., where this feature may result in exposed or shorted conductors.
- Harness routing should not be contact sharp edges (e.g., screw threads, that may damage wiring.
- Plastic or rubber grommets over sharp edges are suggested
- A wear buffer of tape or heat shrink over the harness near the sharp edge can protect the harness from chafing.
- Vinyl electrical or Kapton tapes may be considered for this use.
- “Expando” sleeving
- All wiring should have strain relief (small amount of slack) to prevent terminations and connections from demating or intermittent connections.
- Wiring should be cut to the required length with allowances for strain relief and “service loops” to mate/demate major assemblies.
- Excess wiring should not be simply bundled within the electronics bay.

B4.3 Component Installation

- Inertial forces, “g-loads”, need to be considered during the mounting of all components. Considerations include:
- Adequate mounting to prevent movement or breakaway

- The mounting orientation should minimize movement and prevent accidental operation (e.g., of switches), during acceleration.
- The mounting orientation should avoid conditions where spring loaded contact may be lost.

NOTE: Remember that a 10-gram device will effectively "weigh" 100 grams (over 3 ounces) during a 10 g acceleration.

- All interconnected items (e.g., barrier blocks, sensors, batteries, actuators, switches, etc.), should be rigidly fixed to the rocket structure so that they cannot move.
- Rigid fixing implies attachment with clamps, threaded fasteners, or a solid glue bond.
- Cable ties, Velcro, and/or tape used alone are not acceptable examples of rigid fixing.
- Switches should be of the type that are not easily operated by handling and/or inertial forces.
- Toggle switches, especially those with "bat" shaped toggles should be avoided.
- Slide switches, if used, should be mounted perpendicular to the direction of flight.
- Rotary or key switches are typically immune to inertial forces and accidental operation.
- Make sure that tools or keys for switches are accounted for in procedures.
- Some rotary switches, e.g. ones used to switch input voltage on AC powered equipment, are not designed for a large number of actuation cycles. While rocket applications are typically not demanding for the number of cycles, having a robust switch design is highly recommended.
- Externally mounted button switches are easy to accidently operate during rocket handling. Place a collar around the switch to help prevent the rocket weight from operating the switch.
- Screw switches are generally immune to inertial forces but may change state from vibration. Slight mutilation of the screw threads with a pair of pliers can increase the forces required to open/close the switch.
- The preferred mounting for roller and leaf type snap action switches (also called microswitches) is horizontal to prevent actuation from inertial forces.
- If using pull pin type switches consider using a secondary restraint (e.g. tape) for the pin to prevent the pin from falling out during transportation.
- If using RCA plugs or barrel/tip connectors (also called coaxial power connectors) as pull pins consider using a secondary restraint (e.g. tape) for the connector to prevent the pin from falling out during transportation.
- Some switches are assembled by crimping components together. Make sure that crimps are robust to withstand flight loads; avoid switches that do not have sufficient crimps.

B4.4 Batteries

- Battery capacity and charging may be adversely affected by temperatures at the IREC. Batteries/airframes stored within vehicles can reach 140ºF.
- Monitor the conditions where batteries are stored.
- Use light color paints for avionics compartments.
- Battery connections should be clearly identified for polarity. This may be accomplished by unambiguous labeling, color coded wiring (typically red for positive and black for negative), and preferably polarized connections.
- Batteries should be mounted and connected appropriately:
- "Velcro" (or similar hook and loop material) and/or tie wraps are not permitted as the sole method of mounting batteries.

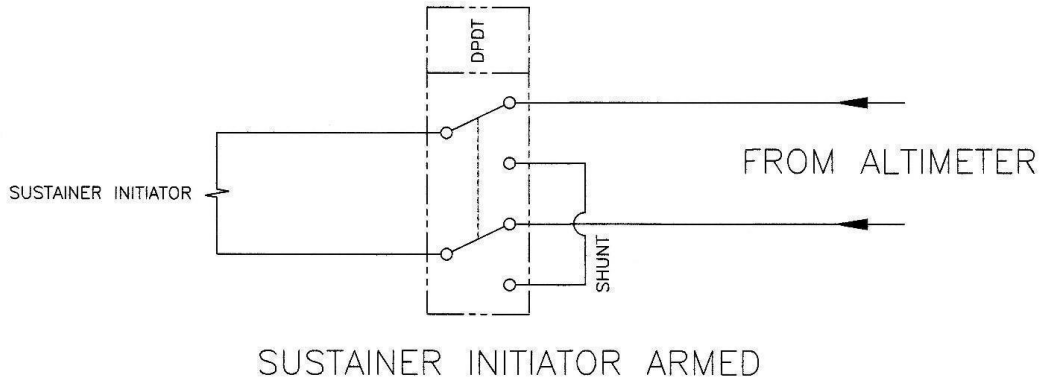
- Batteries vulnerable to puncture damage (if permitted) should be mounted in a fashion and location to mitigate the likelihood of such damage during a hard landing or other high impact event.
- Verify that "coin cell" battery holders typically used on circuit boards adequately retain the battery within the holder.
- Verify that the battery holder is adequately secured to the circuit board to resist inertial forces.
- Rectangular 9V batteries will be secured to prevent motion and connected using proper snap terminals.
- Batteries will be mounted in a direction and manner that prevents acceleration forces from disconnecting the battery connector.
- Cylindrical batteries (AAA, AA, C, D, 18650, etc.) may be mounted into commercial holders.
- Soldering directly to cylindrical batteries is not permitted. Spot welded solder tabs may be used to solder connections to cylindrical batteries.
- Battery holders, if used, will be rigidly secured to the structure, and the batteries will then be strapped (e.g., tie wrap, lacing) into the holders.
- Battery holders will be mounted in a direction and manner that prevents acceleration forces from compressing contact springs.

NOTE: Battery holders should typically be oriented perpendicular to the direction of flight.

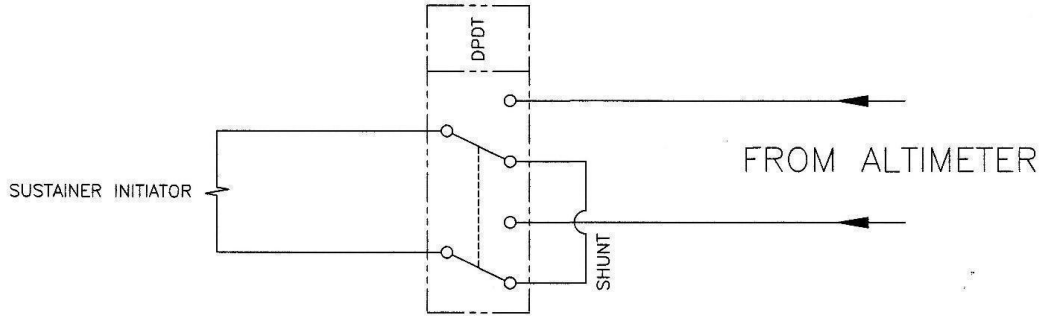
B4.5 Circuit Board Guidelines

- Solderless breadboards (also known as plug-in breadboards) may not be used.
- "Project" type circuit board (e.g., "Vector" boards), that require soldered mounting of components and interconnects may be used.
- Wire wrap, through-hole solder, and surface-mount solder are acceptable fabrication methods.
- Circuit boards will be mounted on standoffs with clearance between the bottom of the board and the mounting structure.
- An adequate number of mounting locations will be used to avoid stressing the circuit board.
- All components greater than 7g will be staked to prevent motion under acceleration.
- The use of conformal coatings is suggested to provide a degree of staking for all components and to avoid electrical shorts.
- All socketed components (e.g., integrated circuits) and press-fit contacts will be positively restrained so that they cannot come loose under vibration.
- The connector between the motherboard and daughterboards will not be the sole means of support.
- Mechanical supports (e.g., fasteners and standoffs), for the daughterboard are required.
- Cuts and jumpers are permitted if the jumpers are fully staked to prevent damage.

APPENDIX C — AIR START IGNITION WIRING DIAGRAM



SUSTAINER INITIATOR ARMED



SUSTAINER INITIATOR SAFE

APPENDIX D — ESRA & TRIPOLI ROCKET ASSOCIATION PARTNERSHIP

Summary

The Experimental Sounding Rocket Association (ESRA) and the Tripoli Rocketry Association (TRA) are now formally working together to continue to improve overall flight safety and efficient flight operations at the annual IREC. Going forward, all operations and all flights at the IREC, whether competitive or demonstration, will be conducted under the TRA Unified Safety Code. Details of how these impacts competing teams are described below.

Background

TRA and ESRA share common goals to create safe and exciting launch opportunities for the next generation of Aerospace engineers as they progress from hobby to industry environments. Our organizations are highly aligned: ESRA provides the educational framework and administration of the world's largest international collegiate rocketry event; and TRA provides the safety/flight operations framework, incredible membership expertise and an amazing insurance program. Stated simply, we have assembled outstanding partners to build and safely operate the greatest rocketry competition in history.

The IREC has relied on the TRA Unified Safety Code and membership expertise/guidance to Create a safe launching environment for its event since 2017. IREC's Range Safety and Launch Operations teams are filled with a significant number of Senior Level 3 high power rocketry experts (>30). A growing number of competing teams utilize TRA prefectures for mentorship, certification flights and test flights of their competition projects.

Benefits:

- **Student Team Liability Insurance** – In previous years, students were either insured by their university or through a 3rd party insurance provider. By becoming a TRA student member and following the TRA Unified Safety Code, the student is insured by TRA's insurance program. TRA insurance covers COTS and SRAD solid and hybrid flights, as well as liquid engine flights that conform to the requirements in the Safety Code.
- Student teams will have improved access to TRA mentors, certifications, and launch sites – TRA has a growing list of US and international prefectures with active launch sites. Student teams are highly recommended to engage with their local TRA prefectures for mentoring, certification flights and additional launch experience.

Tripoli Flight Requirements for All Competition Categories:

- A TRA-certified Level 2 or Level 3 Flyer of Record (depending upon motor impulse) shall be required for all competition categories and must be present for launch preparation, pad loading and recovery activities.[19] There are three options to satisfy the requirement for a Level 2/3 certified Flyer of Record.
    - Recommended: Each team will secure a TRA Level 2/3 certified Adult Flier who works closely with the team and is present as the Flyer of Record for the launch preparation, pad loading and recovery activities. Student teams ARE ENOURAGED to subsidize the travel expenses of their Flyer of Record, both to and from the event.

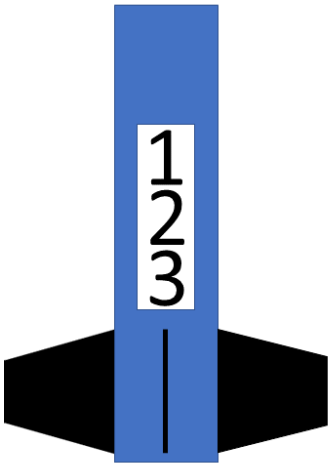
[19] As described in Section 3.1, a FoR with a Level 3 certification is required for flights that include an M, N or O motor or a combination of motors exceeding a total impulse of 5,120Ns. A FoR with a Level 2 certification is acceptable for flights that include a K or L motor or a combination of motors having a total impulse of less than 5,120Ns.

    - Also recommended: A student, who is TRA-certified Adult Flier, and who has significant rocketry experience, can serve as the FoR. The student FoR must be onsite with the team for all aspects of the review and launch.
    - International Teams should begin their FoR search immediately and utilize social media platforms, forums and other community outlets to secure a Flyer of Record. ESRA WILL NOT BE PROVIDING L2 or L3 FoR's.
- TRA membership is required to be on the range or to work on safety-related systems on the rocket.
- Motors or energetic materials may only be possessed or handled by TRA members with appropriate high power rocketry certification.
- Student teams are required to meet all TRA launch safety requirements to ensure TRA insurance covers their flight and their participation at the IREC.

TRA Student Membership

- TRA Student membership is $20 per year per student. This is a requirement for ANY student who will be at the launch site.
- A maximum of 5 additional team members who are TRA student members may be on the pad loading team. These student team members do not have to be HPR certified but must be TRA student members. All students are encouraged to secure HPR certification.

APPENDIX E — EXAMPLE BOOSTER AIRFRAME MARKINGS



IREC Team ID Number 123 on a 4 fin, 6-inch diameter rocket

APPENDIX F — DOCUMENT REVISIONS

REVISION	DESCRIPTION	DATE
2027 REV P1	Minor Revision for 2027 IREC Competition There are numerous edits throughout the document intended to improve the clarity of the document. ESRA recommends that teams carefully review any sections that may affect their participation in the 2027 IREC. Unless otherwise stated, listed section numbers refer to the section numbers in this document and not the prior version. **Added 4.2.1**. New listings are provided for recommended COTS GPS systems. **Modify 4.3**. General clarifications and additions related to the anticipated frequency allocation procedures. **Modify 4.3.3**. Requires that teams will need to design their radio systems such that frequency changes can be made quickly without the need to remove shear pins or access the interior of the rocket. **Modified 4.5.1**. Allows teams to use the HAM License of a Mentor, FoR or ESRA Volunteer. **Added 4.5.4**. Requires a HAM License, and the display of the Call Sign, for any video transmission from the rocket. **Clarify 5.16.2**. Clarifies that a successful SRAD motor test mut be completed, with results submitted for review by the ESRA safety reviewer, prior to submitting the Technical Report. **Added 5.5.4**. Specifies that the use of ammonium nitrate in SRAD motors is prohibited. **Clarify 6.11.6**. Clarifies the procedure for allowing the use of SRAD flight computers for recovery system deployment functions. **Clarify 6.15.6**. Clarifies (with footnote) that teams must be able to safety reach arming switches using either ESRA or team-provided infrastructure. **Modify 7.3.1**. Remove the requirement to deactivate active controls if the rocket tilt exceeds 30%. **Modify 7.4**. Removes the former Section 7.4, which required boost phase dormancy of active controls.	10/15/2027

REVISION	DESCRIPTION	DATE
	Modify 8.2.2. Modification requires documentation of calculated fin flutter velocity but removes the requirement for the fin flutter velocity to be equal or greater than the expected maximum flight velocity. **Clarify 8.3.1**. Clarifies (with footnote) prohibitions on the use of metal in rockets. **Added 8.5.1**. Allows a sliding coupler length of less than 1 caliber to accommodate the use of the so-called head end deployment approach. **Modify 11.2.1**. Requires a justification for the use of a team-provided launch pad. **Modify 11.3**. Significant revisions to the launch control systems for Hybrid and Liquid Rockets. **Modify Appendix B**. This version of the DTEG contains a revised version of Appendix B avionics wiring guidelines. **Remove Appendix C**. Removes the former Appendix C, which provided guidelines for student launch systems. This information is now contained in Section 11. **Remove Appendix G**. Removes the former Appendix G, which listed launch site elevation and weather conditions (because as of the time this version of the DTEG was completed, a site for the 2027 competition has not yet been selected).	